# Masked observations

## 1 Defining equations

**Lemma 1.1.** *For every natural number $Q$ and natural number $t$,* $\mathrm{centergate}(Q, t) = \mathrm{not}(\mathrm{center}(t))\,\mathrm{and}\,\mathrm{rightdefect}(Q, t, 1)$.

*Proof.* The goal follows from $\mathrm{centergatedefinition}(Q, t)$. □

**Lemma 1.2.** *For every natural number $Q$ and natural number $t$,* $\mathrm{alignedbackground}(Q, t) = \mathrm{wittshl}(\mathrm{wittrow}(t), Q)$.

*Proof.* The goal follows from $\mathrm{backgrounddefinition}(Q, t)$. □

**Lemma 1.3.** *For every natural number $Q$ and natural number $t$,* $\mathrm{aligneddefect}(Q, t) = \mathrm{wittxor}(\mathrm{wittrow}(t + Q), \mathrm{alignedbackground}(Q, t))$.

*Proof.* The goal follows from $\mathrm{aligneddefectdefinition}(Q, t)$. □

**Lemma 1.4.** *For every natural number $L$ and natural number $H$,* $\mathrm{carryledger}(L, H, 0) = 0$.

*Proof.* The goal follows from $\mathrm{carryledgerdefinition}(L, H)$. □

**Lemma 1.5.** *For every natural number $L$ and natural number $H$ and natural number $t$,* $\mathrm{carryledger}(L, H, \mathrm{succ}(t)) = \mathrm{wittxor}(\mathrm{carryledger}(L, H, t), \mathrm{wittxor}(\mathrm{wittcarry}(\mathrm{page}(L, H, t)), \mathrm{na}$

*Proof.* The goal follows from $\mathrm{carryledgerdefinition2}(L, H, t)$. □

**Lemma 1.6.** *For every natural number $n$,* $\mathrm{centerflipmask}(n) = \mathrm{wittshl}(7, n)$.

*Proof.* The goal follows from $\mathrm{centerflipmaskdefinition}(n)$. □

**Lemma 1.7.** *For every natural number $start$,* $\mathrm{centersign}(start, 0) = 1$.

*Proof.* The goal follows from $\mathrm{centersigndefinition}(start)$. □

**Lemma 1.8.** *For every natural number $start$ and natural number $t$,* $\mathrm{centersign}(start, \mathrm{succ}(t)) = (1 - 2 \cdot \mathrm{bitvalue}(\mathrm{center}(start))) \cdot \mathrm{centersign}(start + 1, t)$.

*Proof.* The goal follows from $\mathrm{centersigndefinition2}(start, t)$. □

**Lemma 1.9.** *For every natural number $b$ and natural number $d$,* $\mathrm{defectstep}(b, d) = \mathrm{wittxor}(\mathrm{wittphi}(d), \mathrm{wittshl}(\mathrm{crosscarry}(b, d), 1))$.

*Proof.* The goal follows from $\mathrm{defectstepdefinition}(b, d)$. □

**Lemma 1.10.** *For every natural number $Q$ and natural number $d$ and natural number $t$,* diagonalphase$(Q, d, t)$ = bitvalue(ldiag$(d, t{+}Q)$)$-$bitvalue(ldiag$(d, t)$).

*Proof.* The goal follows from diagonalphasedefinition$(Q, d, t)$. □

**Lemma 1.11.** *For every natural number $Q$ and natural number $t$,* observersourceflux$(Q, t, 0)$ = 0.

*Proof.* The goal follows from observersourcefluxdefinition$(Q, t)$. □

**Lemma 1.12.** *For every natural number $Q$ and natural number $t$ and natural number $t1$,* observersourceflux$(Q, t, \text{succ}(t1))$ = wittxor(wittlinear(observersourceflux$(Q, t, t1)$), wittshl($t1), t1 + 1)$).

*Proof.* The goal follows from observersourcefluxdefinition2$(Q, t, t1)$. □

**Lemma 1.13.** *For every natural number $L$ and natural number $H$,* pageboundarycarry$(L, H, 0)$ = false.

*Proof.* The goal follows from pageboundarycarrydefinition$(L, H)$. □

**Lemma 1.14.** *For every natural number $L$ and natural number $H$ and natural number $t$,* pageboundarycarry$(L, H, \text{succ}(t))$ = rowbit(page$(L, H, t + 1), 0)$ and rowbit(page$(L, H, t), L - 1)$.

*Proof.* The goal follows from pageboundarycarrydefinition2$(L, H, t)$. □

**Lemma 1.15.** *For every natural number $L$ and natural number $H$,* pagefold$(L, H, 0)$ = 0.

*Proof.* The goal follows from pagefolddefinition$(L, H)$. □

**Lemma 1.16.** *For every natural number $L$ and natural number $H$ and natural number $t$,* pagefold$(L, H, \text{succ}(t))$ = wittxor(pagefold$(L, H, t)$, page$(L, H, t)$).

*Proof.* The goal follows from pagefolddefinition2$(L, H, t)$. □

**Lemma 1.17.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* pathflux$(Q, t, k, 0) = 0$.

*Proof.* The goal follows from pathfluxdefinition$(Q, t, k)$. □

**Lemma 1.18.** *For every natural number $d$,* qordersearch$(d, 0) = 0$.

*Proof.* The goal follows from qordersearchdefinition$(d)$. □

**Lemma 1.19.** *For every natural number $t$ and natural number $k$,* $\mathrm{rayflux}(t, k, 0) = 0$.

*Proof.* The goal follows from $\mathrm{rayfluxdefinition}(t, k)$. □

**Lemma 1.20.** *For every truth value $a$ and truth value $g$ and integer $outgoing$ and integer $gate$,* $\mathrm{recoverphase}(a, g, outgoing, gate) = (1 - 2 \cdot \mathrm{bitvalue}(g)) \cdot (outgoing - (1 - 2 \cdot \mathrm{bitvalue}(a)) \cdot gate)$.

*Proof.* The goal follows from $\mathrm{recoverphasedefinition}(a, g, outgoing, gate)$. □

**Lemma 1.21.** *For every natural number $t$,* $\mathrm{rightimport}(t) = \mathrm{not}(\mathrm{rightcell}(t, 1))\,\mathrm{and}\,\mathrm{rightcell}(t, 2)$.

*Proof.* The goal follows from $\mathrm{rightimportdefinition}(t)$. □

**Lemma 1.22.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* $\mathrm{rightphase}(Q, t, k) = \mathrm{bitvalue}(\mathrm{rightcell}(t{+}Q, k)){-}\mathrm{bitvalue}(\mathrm{rightcell}(t, k))$.

*Proof.* The goal follows from $\mathrm{rightphasedefinition}(Q, t, k)$. □

**Lemma 1.23.** *For every truth value $a$ and truth value $g$ and integer $incoming$ and integer $gate$,* $\mathrm{transferphase}(a, g, incoming, gate) = (1 - 2 \cdot \mathrm{bitvalue}(g)) \cdot incoming + (1 - 2 \cdot \mathrm{bitvalue}(a)) \cdot gate$.

*Proof.* The goal follows from $\mathrm{transferphasedefinition}(a, g, incoming, gate)$. □

**Lemma 1.24.** *For every natural number $n$,* $\mathrm{twopow}(\mathrm{succ}(n)) = \mathrm{twopow}(n){+}\,\mathrm{twopow}(n)$.

*Proof.* The goal follows from $\mathrm{twopowdefinition2}(n)$. □

**Lemma 1.25.** *For every natural number $y$,* $\mathrm{wittlinear}(y) = \mathrm{wittxor}(\mathrm{wittxor}(y, \mathrm{wittshl}(y, 1)), \mathrm{wittshl}(y, 2$

*Proof.* The goal follows from $\mathrm{philineardefinition}(y)$. □

**Lemma 1.26.** *For every natural number $h1$ and natural number $m$,* $\mathrm{wittlinpow}(\mathrm{succ}(m), h1) = \mathrm{wittlinear}(\mathrm{wittlinpow}(m, h1))$.

*Proof.* The goal follows from $\mathrm{philinearpowdefinition2}(h1, m)$. □

**Lemma 1.27.** *For every natural number $n$,* $\mathrm{wittrow}(\mathrm{succ}(n)) = \mathrm{wittphi}(\mathrm{wittrow}(n))$.

*Proof.* The goal follows from $\mathrm{rowdefinition2}(n)$. □

**Lemma 1.28.** *For every natural number $Q$ and natural number $t$ and natural number $i$,* $\mathrm{alignedphase}(Q, t, i) = \mathrm{pairphase}(\mathrm{alignedbackground}(Q, t), \mathrm{aligneddefect}(Q, t), i)$.

*Proof.* The goal follows from alignedphasedefinition$(Q, t, i)$. □

**Lemma 1.29.** *For every natural number $h$ and natural number $A$,* blockendpoint$(h, A) =$ rowbit$(A, 0)$ and rowbit$(A, h - 1)$.

*Proof.* The goal follows from blockendpointdefinition$(h, A)$. □

**Lemma 1.30.** *For every natural number $t$,* center$(t) =$ cell$(t, 0)$.

*Proof.* The goal follows from centerdefinition$(t)$. □

**Lemma 1.31.** *For every natural number $Q$ and natural number $d$ and natural number $t$ and natural number $p$,* diagonalflux$(Q, d, t, p) =$ rayflux$(t + Q, t + Q - d, t + Q - d - p) -$ rayflux$(t, t - d, t - d - p)$.

*Proof.* The goal follows from diagonalfluxdefinition$(Q, d, t, p)$. □

**Lemma 1.32.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* erasureevent$(Q, t, k) =$ (leftdefect$(Q, t, k+1) =$ true $\wedge$ leftdefect$(Q, t+1, k) =$ false $\wedge$ gatecode$(Q, t, k) \neq 0 \wedge$collisioncode$(Q, t, k) = 0 \wedge$(leftcolour$(Q, t+1, k) = 0 \vee$ leftcolour$(Q, t + 1, k) = 3)$).

*Proof.* The goal follows from erasureeventdefinition$(Q, t, k)$. □

**Lemma 1.33.** ldiag$(0, 0) =$ true.

*Proof.* The goal follows from ldiagdefinition. □

**Lemma 1.34.** *For every natural number $n$,* ldiag$(\text{succ}(n), 0) =$ false.

*Proof.* The goal follows from ldiagdefinition2$(n)$. □

**Lemma 1.35.** *For every natural number $s$,* ldiag$(0, \text{succ}(s)) =$ true.

*Proof.* The goal follows from ldiagdefinition3$(s)$. □

**Lemma 1.36.** *For every natural number $s$,* ldiag$(\text{succ}(0), \text{succ}(s)) =$ true.

*Proof.* The goal follows from ldiagdefinition4$(s)$. □

**Lemma 1.37.** *For every natural number $s$ and natural number $n1$,* ldiag$(\text{succ}(\text{succ}(n1)), \text{succ}(s)) =$ bxor(ldiag$(n1, s)$, ldiag$(\text{succ}(n1), s)$ or ldiag$(\text{succ}(\text{succ}(n1)), s)$).

*Proof.* The goal follows from ldiagdefinition5$(s, n1)$. □

**Lemma 1.38.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* $\mathrm{leadingadvance}(Q, t, k) = (2 \leq k \wedge \mathrm{advancingatdepth}(Q, t, k) \wedge \mathrm{rightmostleft}(Q, t, k))$.

*Proof.* The goal follows from $\mathrm{leadingadvancedefinition}(Q, t, k)$. □

**Lemma 1.39.** *For every truth value $b$,* $\mathrm{naturalbit}(b) = \mathrm{naturalbit}(b)$.

*Proof.* The goal follows from $\mathrm{naturalbitdefinition}(b)$. □

**Lemma 1.40.** *For every natural number $t$,* $\mathrm{observerinput}(t) = \mathrm{not}(\mathrm{center}(t))$ and $\mathrm{rightcell}(t, 1)$.

*Proof.* The goal follows from $\mathrm{observerinputdefinition}(t)$. □

**Lemma 1.41.** *For every natural number $Q$ and natural number $t$,* $\mathrm{observerqrank}(Q, t) = t + Q - 2 \cdot \mathrm{qorder}(\mathrm{observerdefect}(Q, t))$.

*Proof.* The goal follows from $\mathrm{observerqrankdefinition}(Q, t)$. □

**Lemma 1.42.** *For every natural number $b$ and natural number $d$ and natural number $i$,* $\mathrm{pairphase}(b, d, i) = \mathrm{bitvalue}(\mathrm{rowbit}(\mathrm{wittxor}(b, d), i)) - \mathrm{bitvalue}(\mathrm{rowbit}(b, i))$.

*Proof.* The goal follows from $\mathrm{pairphasedefinition}(b, d, i)$. □

**Lemma 1.43.** *For every natural number $a$ and natural number $i$,* $\mathrm{rowbit}(a, i) = \mathrm{rowbit}(a, i)$.

*Proof.* The goal follows from $\mathrm{rowbitdefinition}(a, i)$. □

**Lemma 1.44.** *For every natural number $h1$,* $\mathrm{wittwin}(h1, 0) = 0$.

*Proof.* The goal follows from $\mathrm{winwittdefinition}(h1)$. □

**Lemma 1.45.** *For every natural number $h1$ and natural number $s$,* $\mathrm{wittwin}(h1, \mathrm{succ}(s)) = \mathrm{wittxor}(\mathrm{wittlinear}(\mathrm{wittwin}(h1, s)), \mathrm{wittshl}(\mathrm{wittcarry}(\mathrm{wittrow}(h1 + s)), 1))$.

*Proof.* The goal follows from $\mathrm{winwittdefinition2}(h1, s)$. □

**Lemma 1.46.** *For every natural number $n$ and natural number $j$ and truth value $b$,* $\mathrm{bitcarryeq}(n, j, b) = (\mathrm{wittbit}(\mathrm{wittcarry}(n), j) = \mathrm{bton}(b))$.

*Proof.* The goal follows from $\mathrm{bitcarryeqdefinition}(n, j, b)$. □

**Lemma 1.47.** *For every natural number $a$ and natural number $b$ and natural number $j$ and truth value $c$,* $\mathrm{bitcrosscarryeq}(a, b, j, c) = (\mathrm{wittbit}(\mathrm{crosscarry}(a, b), j) = \mathrm{bton}(c))$.

*Proof.* The goal follows from bitcrosscarryeqdefinition$(a, b, j, c)$. □

**Lemma 1.48.** *For every natural number $a$ and natural number $b$ and natural number $j1$ and natural number $j2$ and truth value $c$,* bitcrosscarryxoreq$(a, b, j1, j2, c) =$ (bxor(rowbit(crosscarry$(a, b), j1)$, rowbit(crosscarry$(a, b), j2)) = c$).

*Proof.* The goal follows from bitcrosscarryxoreqdefinition$(a, b, j1, j2, c)$. □

**Lemma 1.49.** *For every natural number $k$ and natural number $y$ and natural number $j$ and truth value $b$,* bitfrobdefecteq$(k, y, j, b) =$ (wittbit(frobdefect$(k, y), j) =$ bton$(b)$).

*Proof.* The goal follows from bitfrobdefecteqdefinition$(k, y, j, b)$. □

**Lemma 1.50.** *For every natural number $t0$ and natural number $m$ and natural number $j$ and truth value $b$,* bitwinwitteq$(t0, m, j, b) =$ (wittbit(wittwin$(t0, m), j) =$ bton$(b)$).

*Proof.* The goal follows from bitwinwitteqdefinition$(t0, m, j, b)$. □

**Lemma 1.51.** *For every natural number $k$ and natural number $y$,* frobdefect$(k, y) =$ wittxor(wittcarry(wittscale$(k, y))$, wittscale$(k,$ wittcarry$(y)))$.

*Proof.* The goal follows from frobdefectdefinition$(k, y)$. □

**Lemma 1.52.** *For every natural number $t0$,* maskcount$(t0, 0) =$ false.

*Proof.* The goal follows from maskcountdefinition$(t0)$. □

**Lemma 1.53.** *For every natural number $t$ and natural number $k$,* maskparity$(t, k) =$ maskcount$(t,$ twopow$(k))$.

*Proof.* The goal follows from maskparitydefinition$(t, k)$. □

**Lemma 1.54.** *For every natural number $t$,* rcenter$(t) =$ rdiag$(t, t)$.

*Proof.* The goal follows from rcenterdefinition$(t)$. □

**Lemma 1.55.** rdiag$(0, 0) =$ true.

*Proof.* The goal follows from rdiagdefinition. □

**Lemma 1.56.** *For every natural number $n$,* rdiag(succ$(n), 0) =$ false.

*Proof.* The goal follows from rdiagdefinition2$(n)$. □

**Lemma 1.57.** *For every natural number $s$,* rdiag$(0,$ succ$(s)) =$ true.

*Proof.* The goal follows from rdiagdefinition3($s$). □

**Lemma 1.58.** *For every natural number $s$,* rdiag(succ(0), succ($s$)) = not(rdiag(succ(0), $s$)).

*Proof.* The goal follows from rdiagdefinition4($s$). □

**Lemma 1.59.** *For every natural number $s$ and natural number $n1$,* rdiag(succ(succ($n1$)), succ($s$)) = bxor(rdiag(succ(succ($n1$)), $s$), rdiag(succ($n1$), $s$) or rdiag($n1$, $s$)).

*Proof.* The goal follows from rdiagdefinition5($s$, $n1$). □

**Lemma 1.60.** *For every natural number $t$ and natural number $k$,* response($t$, $k$) = window($t$, twopow($k$), 0).

*Proof.* The goal follows from responsedefinition($t$, $k$). □

**Lemma 1.61.** *For every truth value $a$ and truth value $b$ and truth value $c$,* rule($a$, $b$, $c$) = bxor($a$, $b$ or $c$).

*Proof.* The goal follows from ruledefinition($a$, $b$, $c$). □

**Lemma 1.62.** twopow(0) = 1.

*Proof.* The goal follows from twopowdefinition. □

**Lemma 1.63.** *For every natural number $h1$ and integer $h2$,* window($h1$, 0, $h2$) = false.

*Proof.* The goal follows from window60definition($h1$, $h2$). □

**Lemma 1.64.** *For every natural number $y$ and natural number $j$,* wittbitbool($y$, $j$) = rowbit($y$, $j$).

*Proof.* The goal follows from wittbitbooldefinition($y$, $j$). □

**Lemma 1.65.** *For every natural number $t$,* wittcenter($t$) = wittbit(wittrow($t$), $t$).

*Proof.* The goal follows from rowcenterdefinition($t$). □

**Lemma 1.66.** *For every natural number $h1$,* wittlinpow(0, $h1$) = $h1$.

*Proof.* The goal follows from philinearpowdefinition($h1$). □

**Lemma 1.67.** *For every natural number $t$ and natural number $k$,* wittresp($t$, $k$) = rowbit(wittwin($t$, twopow($k$)), $t$ + twopow($k$)).

*Proof.* The goal follows from wittrespbooldefinition($t$, $k$). □

**Lemma 1.68.** wittrow$(0) = 1$.

*Proof.* The goal follows from rowdefinition. □

**Lemma 1.69.** *For every natural number $k$ and natural number $y$,* wittscale$(k, y) =$ wittxor(wittxor$(y,$ wittshl$(y,$ twopow$(k))),$ wittshl$(y,$ twopow$(k)+$twopow$(k)))$.

*Proof.* The goal follows from phiscaledefinition$(k, y)$. □

**Lemma 1.70.** *For every natural number $t$,* ledgerdiagonal$(t) =$ rowbit(wittwin$(0, t), t)$.

*Proof.* The goal follows from ledgerdiagonaldefinition$(t)$. □

**Lemma 1.71.** *For every natural number $t$ and natural number $j$,* ledgerbit$(t, j) =$ rowbit(wittwin$(0, t), j)$.

*Proof.* The goal follows from ledgerbitdefinition$(t, j)$. □

## 2 Masked observations

**Theorem 2.1.** *For every natural number $t$,* whitecount$(t, 0) = 0$.

*Proof.* The goal follows from whitecountzero$(t)$. □

**Theorem 2.2.** *For every natural number $t$ and natural number $n$, if* center$(t) =$ true*, then* whitecount$(t, n + 1) =$ whitecount$(t + 1, n)$.

*Proof.* The goal follows from whitecountblackstep$(t, n)$. □

**Theorem 2.3.** *For every natural number $t$ and natural number $n$, if* center$(t) =$ false*, then* whitecount$(t, n + 1) =$ whitecount$(t + 1, n) + 1$.

*Proof.* The goal follows from whitecountwhitestep$(t, n)$. □

**Theorem 2.4.** *For every natural number $t1$ and natural number $t2$ and natural number $n$, if* centerwindowagree$(t1, t2, n)$*, then* leftprefixagree$(t1, t2, n)$ *if and only if* whitetraceagree$(t1, t2, n)$.

*Proof.* The goal follows from prefixagreementiffwhitetrace$(t1, t2, n)$. □

**Theorem 2.5.** *For every natural number $T$ and natural number $P$ and natural number $t$ and natural number $n$, if $0 < P$, then if* centerperiodic$(T, P)$*, then if $T \leq t$, then there exists a natural number $a$ such that there exists a natural number $b$ such that $a < b$ and $b \leq$* twopow(whitecount$(t, n)) \wedge$ leftprefixagree$(t + a \cdot P, t + b \cdot P, n)$.

*Proof.* The goal follows from compressedprefixrepeat$(T, P, t, n)$. □

**Theorem 2.6.** *For every natural number $T$ and natural number $P$ and natural number $t$ and natural number $n$ and natural number $M$, if $0 < P$, then if* centerperiodic$(T, P)$*, then if $T \leq t$, then if* prefixseparated$(t, P, n, M)$*, then* $M \leq$ twopow(whitecount$(t, n)$).

*Proof.* The goal follows from separatedprefixbound$(T, P, t, n, M)$. □

**Theorem 2.7.** *For every natural number $T$ and natural number $P$ and natural number $t$ and natural number $n$, if $0 < P$, then if* centerperiodic$(T, P)$*, then if $T \leq t$, then* $n < t +$ twopow(whitecount$(t, n)) \cdot P$.

*Proof.* The goal follows from edgeinformationbound$(T, P, t, n)$. □

**Theorem 2.8.** *For every natural number $n$ and natural number $t$ and natural number $w$ and natural number $P$, if $n < (w+1) \cdot P$, then $n < t+$*twopow$(w) \cdot P$.

*Proof.* The goal follows from informationboundaudit$(n, t, w, P)$. □

## 3 Central collision recurrence

**Theorem 3.1.** *For every natural number $Q$ and natural number $t$ and natural number $r$, if* leftadjacentdefect$(Q, t) =$ true*, then if* center$(t + r) =$ true*, then if* center$(t + r + Q) =$ true*, then* nearcollision$(Q, t, r + 1, 2 \cdot (r + 1))$.

*Proof.* The goal follows from unitdefectforcesnearcollision$(Q, t, r)$. □

**Theorem 3.2.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq t$, then if* leftadjacentdefect$(m \cdot P, t) =$ true*, then* nearcollision$(m \cdot P, t, P, 2 \cdot P)$.

*Proof.* The goal follows from periodicunitdefectforcesnearcollision$(T, P, m, t)$. □

**Theorem 3.3.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq N$, then there exists a natural number $t$ such that $N \leq t$ and $t+1 \leq 2 \cdot N + m \cdot P + P \wedge$*centralcollision$(m \cdot P, t, 2 \cdot P)$.

*Proof.* The goal follows from periodiccentralcollisionwindow$(T, P, m, N)$. □

**Theorem 3.4.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then there exists a natural number $t$ such that $N \leq t$ and* centralcollision$(m \cdot P, t, 2 \cdot P)$*.*

*Proof.* The goal follows from periodiccentralcollisions$(T, P, m, N)$. □

**Theorem 3.5.** *For every natural number $T$ and natural number $P$ and natural number $m$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then* recurrentcentralcollision$(m \cdot P, 2 \cdot P)$*.*

*Proof.* The goal follows from periodicrecurrentcentralcolumn$(T, P, m)$. □

**Theorem 3.6.** *For every natural number $Q$ and natural number $b$ and natural number $r$, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) =$ true*, then if* leftadjacentdefect$(Q, b+r) =$ true*, then there exists a natural number $s$ such that $2 \cdot s \leq r+1$ and $s \leq r \wedge$*leftcollision$(Q, b+s, r+1)$*.*

*Proof.* The goal follows from blackwindowforcesleftcollision$(Q, b, r)$. □

**Theorem 3.7.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq N$, then there exists a natural number $t$ such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge$* leftcollision$(m \cdot P, t, P)$*.*

*Proof.* The goal follows from periodicleftcollisionwindow$(T, P, m, N)$. □

**Theorem 3.8.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then there exists a natural number $t$ such that $N \leq t$ and* leftcollision$(m \cdot P, t, P)$*.*

*Proof.* The goal follows from periodicleftcollisions$(T, P, m, N)$. □

**Theorem 3.9.** *For every natural number $T$ and natural number $P$ and natural number $m$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then* recurrentleftcollision$(m \cdot P, P)$*.*

*Proof.* The goal follows from periodicrecurrentleftcolumn$(T, P, m)$. □

**Theorem 3.10.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* collisioncode$(Q, t, k) \neq 0$ *if and only if* collisionbit$(Q, t, k) =$ true*.*

*Proof.* The goal follows from collisioncodenonzero$(Q, t, k)$. □

**Theorem 3.11.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* $\mathrm{collisioncode}(Q, t, k) \leq 6$.

*Proof.* The goal follows from $\mathrm{collisioncodebound}(Q, t, k)$. □

**Theorem 3.12.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $0 < k$, *then* $\mathrm{collisioncode}(Q, t, k) = 1 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 1 \wedge \mathrm{leftcolour}(Q, t, k-1) = 2$ *and* $(\mathrm{collisioncode}(Q, t, k) = 2 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 2 \wedge \mathrm{leftcolour}(Q, t, k-1) = 1) \wedge (\mathrm{collisioncode}(Q, t, k) = 3 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 1 \wedge \mathrm{leftcolour}(Q, t, k-1) = 3) \wedge (\mathrm{collisioncode}(Q, t, k) = 4 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 3 \wedge \mathrm{leftcolour}(Q, t, k-1) = 1) \wedge (\mathrm{collisioncode}(Q, t, k) = 5 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 2 \wedge \mathrm{leftcolour}(Q, t, k-1) = 3) \wedge (\mathrm{collisioncode}(Q, t, k) = 6 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 3 \wedge \mathrm{leftcolour}(Q, t, k-1) = 2)$.

*Proof.* The goal follows from $\mathrm{collisioncodecolours}(Q, t, k)$. □

**Theorem 3.13.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* $\mathrm{advancingatdepth}(Q, t, k)$ *if and only if* $\mathrm{collisioncode}(Q, t, k) = 3$ *or* $\mathrm{collisioncode}(Q, t, k) = 5$.

*Proof.* The goal follows from $\mathrm{advancingcollisioncodes}(Q, t, k)$. □

**Theorem 3.14.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\mathrm{collisioncode}(Q, t, k) \neq 0$, *then* $\mathrm{leftdefect}(Q, t+1, k) = \mathrm{leftdefect}(Q, t, k+1)$.

*Proof.* The goal follows from $\mathrm{collisiontransmitsleft}(Q, t, k)$. □

**Theorem 3.15.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\mathrm{collisioncode}(Q, t, k) \neq 0$, *then* $\mathrm{leftgatedifference}(Q, t, k) = \mathrm{false}$.

*Proof.* The goal follows from $\mathrm{collisiongatesagree}(Q, t, k)$. □

**Theorem 3.16.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $2 \leq k$, *then* $\mathrm{collisioncode}(Q, t, k) = 1 \rightarrow \mathrm{leftdefect}(Q, t+1, k-1) = \mathrm{leftcell}(t, k-2)$ *and* $(\mathrm{collisioncode}(Q, t, k) = 2 \rightarrow \mathrm{leftdefect}(Q, t+1, k-1) = \mathrm{leftcell}(t+Q, k-2)) \wedge (\mathrm{collisioncode}(Q, t, k) = 3 \rightarrow \mathrm{leftdefect}(Q, t+1, k-1) = \mathrm{true}) \wedge (\mathrm{collisioncode}(Q, t, k) = 4 \rightarrow \mathrm{leftdefect}(Q, t+1, k-1) = \mathrm{not}(\mathrm{leftcell}(t+Q, k-2))) \wedge (\mathrm{collisioncode}(Q, t, k) = 5 \rightarrow \mathrm{leftdefect}(Q, t+1, k-1) = \mathrm{true}) \wedge (\mathrm{collisioncode}(Q, t, k) = 6 \rightarrow \mathrm{leftdefect}(Q, t+1, k-1) = \mathrm{not}(\mathrm{leftcell}(t, k-2)))$.

*Proof.* The goal follows from $\mathrm{collisionrighttransporttable}(Q, t, k)$. □

**Theorem 3.17.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $0 < k$, then* $\mathrm{collisioncode}(Q, t, k) = 3 \to \mathrm{leftcolour}(Q, t + 1, k-1) = 2$ *and if* $\mathrm{collisioncode}(Q, t, k) = 5$*, then* $\mathrm{leftcolour}(Q, t+1, k-1) = 1$.

*Proof.* The goal follows from $\mathrm{advancingrightcolourtable}(Q, t, k)$. □

**Theorem 3.18.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* $\mathrm{leftdefect}(Q, t + 1, k) = \mathrm{not}(\mathrm{leftdefect}(Q, t, k + 1))$ *if and only if* $\mathrm{leftgatedifference}(Q, t, k) = \mathrm{true}$.

*Proof.* The goal follows from $\mathrm{gatedifferenceiffflip}(Q, t, k)$. □

**Theorem 3.19.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\mathrm{leftdefect}(Q, t, k + 1) = \mathrm{true}$*, then* $\mathrm{leftdefect}(Q, t + 1, k) = \mathrm{false}$ *if and only if* $\mathrm{leftgatedifference}(Q, t, k) = \mathrm{true}$.

*Proof.* The goal follows from $\mathrm{incomingdefecterasediff}(Q, t, k)$. □

**Theorem 3.20.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\mathrm{leftdefect}(Q, t, k + 1) = \mathrm{true}$*, then if* $\mathrm{leftdefect}(Q, t + 1, k) = \mathrm{false}$*, then* $\mathrm{collisioncode}(Q, t, k) = 0$.

*Proof.* The goal follows from $\mathrm{erasurehasnocollision}(Q, t, k)$. □

**Theorem 3.21.** *For every natural number $Q$ and natural number $k$, if $0 < k$, then* $\mathrm{leftcolour}(Q, 0, k) = 0$ *or* $\mathrm{leftcolour}(Q, 0, k) = 2$.

*Proof.* The goal follows from $\mathrm{seedleftcolours}(Q, k)$. □

**Theorem 3.22.** *For every natural number $Q$ and natural number $k$, if $2 \leq k$, then* $\mathrm{collisioncode}(Q, 0, k) = 0$.

*Proof.* The goal follows from $\mathrm{seedleftcollisionzero}(Q, k)$. □

**Theorem 3.23.** $\mathrm{collisioncode}(1, 1, 2) = 5$.

*Proof.* The goal follows from seedadvancingcollisionbirth. □

**Theorem 3.24.** $\mathrm{leftdefect}(3, 0, 3) = \mathrm{true}$ *and* $\mathrm{leftdefect}(3, 1, 2) = \mathrm{false} \wedge \mathrm{leftgatedifference}(3, 0, 2) = \mathrm{true}$.

*Proof.* The goal follows from seedgateerasure. □

**Theorem 3.25.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\mathrm{leftdefect}(Q, t, k) = \mathrm{true}$*, then if* $\mathrm{center}(t) = \mathrm{true}$*, then if* $\mathrm{center}(t + Q) = \mathrm{true}$*, then* $\mathrm{advancingleftcone}(Q, t, k)$.

*Proof.* The goal follows from $\mathrm{leftdefectblackcone}(Q, t, k)$. □

**Theorem 3.26.** *For every natural number $Q$ and natural number $b$ and natural number $r$, if $0 < r$, then if* $\mathrm{centerwindowagree}(b, b + Q, r + 1)$*, then if* $\mathrm{center}(b) = \mathrm{true}$*, then if* $\mathrm{leftadjacentdefect}(Q, b+r) = \mathrm{true}$*, then there exists a natural number $s$ such that $2 \cdot s \leq r+1$ and $s \leq r \wedge$* $\mathrm{advancingleftcollision}(Q, b+s, r+1)$.

*Proof.* The goal follows from $\mathrm{blackwindowforcesadvancingcollision}(Q, b, r)$. □

**Theorem 3.27.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq N$, then there exists a natural number $t$ such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge$* $\mathrm{advancingleftcollision}(m \cdot P, t, P)$.

*Proof.* The goal follows from $\mathrm{periodicadvancingleftcollisionwindow}(T, P, m, N)$. □

**Theorem 3.28.** *For every natural number $T$ and natural number $P$ and natural number $m$, if $0 < P$, then if $0 < m$, then if* $\mathrm{centerperiodic}(T, P)$*, then* $\mathrm{recurrentadvancingleftcollision}(m \cdot P, P)$.

*Proof.* The goal follows from $\mathrm{periodicrecurrentadvancingleftcolumn}(T, P, m)$. □

**Theorem 3.29.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* $\mathrm{leftdefect}(Q, t, k) = \mathrm{false}$ *if and only if* $\mathrm{leftcolour}(Q, t, k) = 0$ *or* $\mathrm{leftcolour}(Q, t, k) = 3$.

*Proof.* The goal follows from $\mathrm{leftdefectfalsecolours}(Q, t, k)$. □

**Theorem 3.30.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* $\mathrm{leftdefect}(Q, t, k) = \mathrm{true}$ *if and only if* $\mathrm{leftcolour}(Q, t, k) = 1$ *or* $\mathrm{leftcolour}(Q, t, k) = 2$.

*Proof.* The goal follows from $\mathrm{leftdefecttruecolours}(Q, t, k)$. □

**Theorem 3.31.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $t < k$, then* $\mathrm{leftcolour}(Q, t, k) = 0$ *or* $\mathrm{leftcolour}(Q, t, k) = 2$.

*Proof.* The goal follows from leftcoloursoutsideseedcone$(Q, t, k)$. □

**Theorem 3.32.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* collisioncode$(Q, t, k) \neq 0$*, then* $k \leq t + 1$.

*Proof.* The goal follows from collisioncodesupport$(Q, t, k)$. □

**Theorem 3.33.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* gatecode$(Q, t, k) \neq 0$ *if and only if* leftgatedifference$(Q, t, k) =$ true.

*Proof.* The goal follows from gatecodenonzero$(Q, t, k)$. □

**Theorem 3.34.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* gatecode$(Q, t, k) \leq 6$.

*Proof.* The goal follows from gatecodebound$(Q, t, k)$. □

**Theorem 3.35.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $0 < k$*, then* gatecode$(Q, t, k) = 1 \leftrightarrow$ leftcolour$(Q, t, k) = 1 \wedge$ leftcolour$(Q, t, k - 1) = 0$ *and* (gatecode$(Q, t, k) = 2 \leftrightarrow$ leftcolour$(Q, t, k) = 0 \wedge$leftcolour$(Q, t, k-1) = 1) \wedge$(gatecode$(Q, t, k) = 3 \leftrightarrow$ leftcolour$(Q, t, k) = 1 \wedge$leftcolour$(Q, t, k-1) = 1) \wedge$(gatecode$(Q, t, k) = 4 \leftrightarrow$ leftcolour$(Q, t, k) = 2 \wedge$leftcolour$(Q, t, k-1) = 0) \wedge$(gatecode$(Q, t, k) = 5 \leftrightarrow$ leftcolour$(Q, t, k) = 0 \wedge$leftcolour$(Q, t, k-1) = 2) \wedge$(gatecode$(Q, t, k) = 6 \leftrightarrow$ leftcolour$(Q, t, k) = 2 \wedge$ leftcolour$(Q, t, k - 1) = 2)$.

*Proof.* The goal follows from gatecodecolours$(Q, t, k)$. □

**Theorem 3.36.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* leftdefect$(Q, t, k + 1) =$ true*, then if* leftdefect$(Q, t + 1, k) =$ false*, then* erasureevent$(Q, t, k)$.

*Proof.* The goal follows from erasureeventofstep$(Q, t, k)$. □

**Theorem 3.37.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* erasureevent$(Q, t, k)$*, then* leftcolour$(Q, t + 1, k) = 3$ *if and only if* gatecode$(Q, t, k) \leq 3$ *and* leftcolour$(Q, t, k + 1) = 2$ *or* $4 \leq$ gatecode$(Q, t, k)$ *and* leftcolour$(Q, t, k + 1) = 1$.

*Proof.* The goal follows from erasureblackphaseiff$(Q, t, k)$. □

**Theorem 3.38.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $t+1 < k$*, then if* erasureevent$(Q, t, k)$*, then* leftcolour$(Q, t+ 1, k) = 0$.

*Proof.* The goal follows from erasureoutsideseedconewhite$(Q, t, k)$. □

**Theorem 3.39.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $2 \leq k$, then if* advancingatdepth$(Q, t, k)$*, then if* leftdefect$(Q, t + k, 0) =$ false*, then there exists a natural number $j$ such that* firstpatherasure$(Q, t, k, j)$.

*Proof.* The goal follows from advancingsourcefirsterasure$(Q, t, k)$. □

**Theorem 3.40.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq N$, then there exists a natural number $u$ such that there exists a natural number $d$ such that $N \leq u$ and $u + 1 \leq 2 \cdot N + m \cdot P + 3 \cdot P \wedge d + 2 \leq P \wedge$* erasureevent$(m \cdot P, u, d)$.

*Proof.* The goal follows from periodicerasurewindow$(T, P, m, N)$. □

**Theorem 3.41.** *For every natural number $T$ and natural number $P$ and natural number $m$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then* recurrentsourceerasure$(m \cdot P, P)$.

*Proof.* The goal follows from periodicrecurrentsourceerasure$(T, P, m)$. □

**Theorem 3.42.** noperiod *if and only if* norecurrentcentralcollision.

*Proof.* The goal follows from noperiodiffnorecurrentcentralcollision. □

**Theorem 3.43.** norowcenterperiod *if and only if* norecurrentcentralcollision.

*Proof.* The goal follows from norowcenterperiodiffnorecurrentcentralcollision. □

**Theorem 3.44.** problemone *if and only if* norecurrentcentralcollision.

*Proof.* The goal follows from problemoneiffnorecurrentcentralcollision. □

**Theorem 3.45.** *If* norecurrentcentralcollision*, then* noperiod.

*Proof.* Assume $h$. The goal follows from noperiodofnorecurrentcentralcollision$(h)$. □

**Theorem 3.46.** *For every natural number $T$ and natural number $P$, if $0 < P$, then if* norecurrentcentralcollision*, then not* centerperiodic$(T, P)$.

*Proof.* The goal follows from notcenterperiodicofnorecurrentcentralcollision$(T, P)$. □

**Theorem 3.47.** *If* norecurrentcentralcollision*, then* problemone*.*

*Proof.* Assume $h$. The goal follows from noteventuallyperiodicofnorecurrentcentralcollision$(h)$. □

**Theorem 3.48.** *For every natural number* $b$ *and natural number* $d$ *and natural number* $i$*,* pairphase(wittphi$(b)$, wittxor(wittphi$(d)$, wittshl(crosscarry$(b, d), 1)), i) =$ bitvalue(rowbit(wittphi(wittxor$(b, d)), i)) -$ bitvalue(rowbit(wittphi$(b), i)$)*.*

*Proof.* The goal follows from pairphasestep$(b, d, i)$. □

**Theorem 3.49.** *For every natural number* $Q$ *and natural number* $t$ *and natural number* $k$*,* alignedphase$(Q, t, t + k + Q) =$ leftphase$(Q, t, k)$*.*

*Proof.* The goal follows from alignedphaseleft$(Q, t, k)$. □

**Theorem 3.50.** *For every natural number* $Q$ *and natural number* $t$ *and natural number* $k$*,* leftphase$(Q, t, k) = 0$ *if and only if* leftdefect$(Q, t, k) =$ false*.*

*Proof.* The goal follows from phasezeroiff$(Q, t, k)$. □

**Theorem 3.51.** *For every natural number* $Q$ *and natural number* $t$ *and natural number* $k$*,* leftphase$(Q, t, k) = -1$ *or* leftphase$(Q, t, k) = 1$ *if and only if* leftdefect$(Q, t, k) =$ true*.*

*Proof.* The goal follows from phaseunitiff$(Q, t, k)$. □

**Theorem 3.52.** *For every natural number* $Q$ *and natural number* $t$ *and natural number* $k$*,* leftphase$(Q, t+1, k) =$ leftphase$(Q, t, k+1)+$leftphaseflux$(Q, t, k)$*.*

*Proof.* The goal follows from leftphasebalance$(Q, t, k)$. □

**Theorem 3.53.** *For every natural number* $Q$ *and natural number* $t$ *and natural number* $k$*, if* collisioncode$(Q, t, k) \neq 0$*, then* leftphaseflux$(Q, t, k) = -2 \cdot$ leftphase$(Q, t, k + 1)$*.*

*Proof.* The goal follows from collisionphaseflip$(Q, t, k)$. □

**Theorem 3.54.** *For every natural number* $Q$ *and natural number* $t$ *and natural number* $k$*, if* $0 < k$*, then if* advancingatdepth$(Q, t, k)$*, then* leftphase$(Q, t+ 1, k - 1) = -$ leftphase$(Q, t, k)$*.*

*Proof.* The goal follows from advancingphaseflip$(Q, t, k)$. □

**Theorem 3.55.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* $\mathrm{gatecode}(Q,t,k) = 1 \to \mathrm{leftgatephase}(Q,t,k) = -1$ *and* $(\mathrm{gatecode}(Q,t,k) = 2 \to \mathrm{leftgatephase}(Q,t,k) = -1) \wedge (\mathrm{gatecode}(Q,t,k) = 3 \to \mathrm{leftgatephase}(Q,t,k) = -1) \wedge (\mathrm{gatecode}(Q,t,k) = 4 \to \mathrm{leftgatephase}(Q,t,k) = 1) \wedge (\mathrm{gatecode}(Q,t,k) = 5 \to \mathrm{leftgatephase}(Q,t,k) = 1) \wedge (\mathrm{gatecode}(Q,t,k) = 6 \to \mathrm{leftgatephase}(Q,t,k) = 1)$.

*Proof.* The goal follows from $\mathrm{gatephasecases}(Q,t,k)$. □

**Theorem 3.56.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\mathrm{erasureevent}(Q,t,k)$, *then* $\mathrm{leftphase}(Q,t+1,k) = 0$ *and* $\mathrm{leftphaseflux}(Q,t,k) = -\mathrm{leftphase}(Q,t,k+1) \wedge (\mathrm{leftphase}(Q,t,k+1) = -1 \vee \mathrm{leftphase}(Q,t,k+1) = 1)$.

*Proof.* The goal follows from $\mathrm{erasurephasebalance}(Q,t,k)$. □

**Theorem 3.57.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\mathrm{erasureevent}(Q,t,k)$, *then* $\mathrm{leftcolour}(Q,t+1,k) = 3 \leftrightarrow \mathrm{leftgatephase}(Q,t,k) \cdot \mathrm{leftphase}(Q,t,k+1) = -1$ *and* $\mathrm{leftcolour}(Q,t+1,k) = 0 \leftrightarrow \mathrm{leftgatephase}(Q,t,k) \cdot \mathrm{leftphase}(Q,t,k+1) = 1$.

*Proof.* The goal follows from $\mathrm{erasureresetproduct}(Q,t,k)$. □

**Theorem 3.58.** *For every natural number $Q$ and natural number $t$ and natural number $k$ and natural number $n$,* $\mathrm{pathphase}(Q,t,k,n) = \mathrm{leftphase}(Q,t,k) + \mathrm{pathflux}(Q,t,k,n)$.

*Proof.* The goal follows from $\mathrm{pathphasebalance}(Q,t,k,n)$. □

**Theorem 3.59.** *For every natural number $Q$ and natural number $t$, if $0 < Q$, then* $\mathrm{leftphase}(Q,t,t+Q) = 1$.

*Proof.* The goal follows from $\mathrm{seedleftphase}(Q,t)$. □

**Theorem 3.60.** *For every natural number $Q$ and natural number $t$, if $0 < Q$, then* $\mathrm{leftphase}(Q,t+(t+Q),0) = 1 + \mathrm{pathflux}(Q,t,t+Q,t+Q)$.

*Proof.* The goal follows from $\mathrm{seedcenterphasebalance}(Q,t)$. □

**Theorem 3.61.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* $\mathrm{centerperiodic}(T,P)$, *then if $T \leq N$, then* $\mathrm{pathflux}(m \cdot P, N, N + m \cdot P, N + m \cdot P) = -1$.

*Proof.* The goal follows from $\mathrm{periodicseedflux}(T,P,m,N)$. □

**Theorem 3.62.** leftphase$(3, 0, 3) = 1$ *and* leftphase$(3, 1, 2) = 0 \wedge$leftphaseflux$(3, 0, 2) = -1 \wedge$ leftgatephase$(3, 0, 2) = 1$.

*Proof.* The goal follows from seedphasecancellation. □

**Theorem 3.63.** *For every natural number $Q$ and natural number $t$ and natural number $k$ and natural number $s$, if* rightmostleft$(Q, t, k)$*, then if $s < k$, then if* centercleanthrough$(Q, t, s)$*, then* rightmostleft$(Q, t + s, k - s)$.

*Proof.* The goal follows from rightmostleftprofile$(Q, t, k, s)$. □

**Theorem 3.64.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* leadingadvance$(Q, t, k)$*, then if* centercleanthrough$(Q, t, k)$*, then* firstpatherasure$(Q, t, k, k - 2)$ *and* erasureevent$(Q, t + (k - 1), 0) \wedge$ (gatecode$(Q, t + (k - 1), 0) = 2 \vee$ gatecode$(Q, t + (k - 1), 0) = 5) \wedge$ center$(t + (k - 1)) =$ false.

*Proof.* The goal follows from leadingfirsterasure$(Q, t, k)$. □

**Theorem 3.65.** *For every natural number $Q$ and natural number $b$ and natural number $r$, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) =$ true*, then if* leftadjacentdefect$(Q, b + r) =$ true*, then there exists a natural number $s$ such that there exists a natural number $k$ such that $2 \cdot s \leq r + 1$ and $s \leq r \wedge k + s \leq r + 1 \wedge$* leadingadvance$(Q, b + s, k)$.

*Proof.* The goal follows from blackwindowforcesleadingsource$(Q, b, r)$. □

**Theorem 3.66.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq N$, then there exists a natural number $t$ such that there exists a natural number $k$ such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge k \leq P \wedge$* leadingadvance$(m \cdot P, t, k)$.

*Proof.* The goal follows from periodicleadingsourcewindow$(T, P, m, N)$. □

**Theorem 3.67.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq N$, then there exists a natural number $t$ such that there exists a natural number $k$ such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge k \leq P \wedge$* leadingadvance$(m \cdot P, t, k) \wedge$ firstpatherasure$(m \cdot P, t, k, k - 2) \wedge$ (gatecode$(m \cdot P, t + (k - 1), 0) = 2 \vee$ gatecode$(m \cdot P, t + (k - 1), 0) = 5)$.

*Proof.* The goal follows from periodicleadingerasurewindow$(T, P, m, N)$. □

**Theorem 3.68.** *For every natural number $T$ and natural number $P$ and natural number $m$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$, *then* recurrentleadingerasure$(m \cdot P, P)$.

*Proof.* The goal follows from periodicrecurrentleadingerasure$(T, P, m)$. □

**Theorem 3.69.** *For every natural number $Q$ and natural number $R$ and natural number $t$ and natural number $k$,* leftphase$(Q+R, t, k) =$ leftphase$(Q, t, k)+$ leftphase$(R, t + Q, k)$.

*Proof.* The goal follows from leftphaselagadd$(Q, R, t, k)$. □

**Theorem 3.70.** *For every natural number $Q$ and natural number $R$ and natural number $t$ and natural number $k$ and natural number $n$,* pathflux$(Q + R, t, k, n) =$ pathflux$(Q, t, k, n) +$ pathflux$(R, t + Q, k, n)$.

*Proof.* The goal follows from pathfluxlagadd$(Q, R, t, k, n)$. □

**Theorem 3.71.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* leadingadvance$(Q, t, k)$, *then if* centercleanthrough$(Q, t, k)$, *then* leftcolour$(Q, t+k, 0) = 3 \leftrightarrow$ leftgatephase$(Q, t+(k-1), 0) \cdot ($leftphase$(Q, t, k)+$ pathflux$(Q, t, k, k-1)) = -1$ *and* leftcolour$(Q, t+k, 0) = 0 \leftrightarrow$ leftgatephase$(Q, t+(k-1), 0) \cdot ($leftphase$(Q, t, k) +$ pathflux$(Q, t, k, k-1)) = 1$.

*Proof.* The goal follows from leadingphasereset$(Q, t, k)$. □

**Theorem 3.72.** *For every natural number $Q$ and natural number $t$, if* leftdefect$(Q, t, 0) =$ false, *then if* leftdefect$(Q, t + 1, 0) =$ false, *then* erasureevent$(Q, t, 0)$ *if and only if* gatecode$(Q, t, 0) = 2$ *or* gatecode$(Q, t, 0) = 5$.

*Proof.* The goal follows from centererasureiffgate$(Q, t)$. □

**Theorem 3.73.** *For every natural number $Q$ and natural number $b$ and natural number $n$ and natural number $i$ and natural number $j$ and natural number $k$, if* commonwhiteblock$(Q, b, n)$, *then if $i \leq j$, then if $j \leq k$, then if $k \leq n$, then if* gatecode$(Q, b + i, 0) = 2$ *or* gatecode$(Q, b + i, 0) = 5$, *then if* gatecode$(Q, b + k, 0) = 2$ *or* gatecode$(Q, b + k, 0) = 5$, *then* gatecode$(Q, b + j, 0) =$ gatecode$(Q, b + i, 0)$.

*Proof.* The goal follows from whiteblockgateconvex$(Q, b, n, i, j, k)$. □

**Theorem 3.74.** *For every natural number $Q$ and natural number $b$ and natural number $n$ and natural number $i$ and natural number $j$ and natural number $k$, if* commonwhiteblock$(Q, b, n)$*, then if* centercleanthrough$(Q, b, n+1)$*, then if $i \leq j$, then if $j \leq k$, then if $k \leq n$, then if* erasureevent$(Q, b+i, 0)$*, then if* erasureevent$(Q, b+k, 0)$*, then* erasureevent$(Q, b+j, 0)$ *and* gatecode$(Q, b+j, 0) =$ gatecode$(Q, b+i, 0)$.

*Proof.* The goal follows from whiteblockerasureconvex$(Q, b, n, i, j, k)$. □

**Theorem 3.75.** leadingadvance$(4, 5, 2)$ *and* centercleanthrough$(4, 5, 2) \wedge$ gatecode$(4, 6, 0) = 5$.

*Proof.* The goal follows from seedleadinggatefive. □

**Theorem 3.76.** leadingadvance$(7, 7, 4)$ *and* centercleanthrough$(7, 7, 4) \wedge$ gatecode$(7, 10, 0) = 2$.

*Proof.* The goal follows from seedleadinggatetwo. □

**Theorem 3.77.** leadingadvance$(11, 4, 4)$ *and* centercleanthrough$(11, 4, 4) \wedge$ gatecode$(11, 7, 0) = 5 \wedge$ leftcolour$(11, 8, 0) = 3$.

*Proof.* The goal follows from seedleadingblackreset. □

**Theorem 3.78.** *For every truth value incomingBit and truth value laterBit and integer incoming and integer gate,* recoverphase$(incomingBit, laterBit,$ transferphase$(incomingBi$ *incoming*.

*Proof.* The goal follows from transferrecover$(incomingBit, laterBit, incoming, gate)$. □

**Theorem 3.79.** *For every truth value incomingBit and truth value laterBit and integer incoming and integer gate,* transferphase$(incomingBit, laterBit, incoming, gate) =$ 0 *if and only if incoming* = recoverphase$(incomingBit, laterBit, 0, gate)$.

*Proof.* The goal follows from transferkernel$(incomingBit, laterBit, incoming, gate)$. □

**Theorem 3.80.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftphase$(Q, t+1, k) =$ transferphase(leftcell$(t, k+1)$, latergate$(Q, t, k)$, leftphase$(Q, t, k+1)$, leftgatephase$(Q, t, k)$).

*Proof.* The goal follows from phasetransfer$(Q, t, k)$. □

**Theorem 3.81.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* $\mathrm{leftphase}(Q, t, k{+}1) = \mathrm{recoverphase}(\mathrm{leftcell}(t, k{+}1), \mathrm{latergate}(Q, t, k), \mathrm{leftphase}(Q, t{+}1, k), \mathrm{leftgatephase}(Q, t, k))$.

*Proof.* The goal follows from $\mathrm{phaserecover}(Q, t, k)$. □

**Theorem 3.82.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\mathrm{erasureevent}(Q, t, k)$*, then not* $\mathrm{leftgatephase}(Q, t, k) = 0$ *and* $\mathrm{leftphase}(Q, t, k{+}1) = \mathrm{recoverphase}(\mathrm{leftcell}(t, k{+}1), \mathrm{latergate}(Q, t, k), 0, \mathrm{leftgatephase}(Q, t, k))$.

*Proof.* The goal follows from $\mathrm{erasurerecovery}(Q, t, k)$. □

**Theorem 3.83.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$ and natural number $g$, if $0 < P$, then if $0 < m$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $g = 2$ or $g = 5$, then not not there exists a natural number $t$ such that $N \leq t$ and* $\mathrm{erasureevent}(m \cdot P, t, 0) \wedge \mathrm{gatecode}(m \cdot P, t, 0) = g$.

*Proof.* The goal follows from $\mathrm{periodicerasureorientation}(T, P, m, N, g)$. □

**Theorem 3.84.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\mathrm{leadingadvance}(Q, t, k)$*, then if* $\mathrm{centercleanthrough}(Q, t, k)$*, then there exists a natural number $s$ such that $s + 1 < k$ and* $\mathrm{center}(t + s) = \mathrm{true}$.

*Proof.* The goal follows from $\mathrm{leadingblackbeforearrival}(Q, t, k)$. □

**Theorem 3.85.** *For every natural number $Q$ and natural number $b$ and natural number $r$, if $0 < r$, then if* $\mathrm{centerwindowagree}(b, b + Q, r + 1)$*, then if* $\mathrm{center}(b) = \mathrm{true}$*, then if* $\mathrm{whitecentergap}(b, r)$*, then if* $\mathrm{leftadjacentdefect}(Q, b{+}r) = \mathrm{true}$*, then there exists a natural number $k$ such that $k \leq r + 1$ and* $\mathrm{leadingadvance}(Q, b, k)$.

*Proof.* The goal follows from $\mathrm{blackgapforcesinitialleading}(Q, b, r)$. □

**Theorem 3.86.** *For every natural number $Q$ and natural number $t$, if* $\mathrm{leadingadvance}(Q, t, 2)$*, then if* $\mathrm{centercleanthrough}(Q, t, 2)$*, then* $\mathrm{center}(t) = \mathrm{true}$.

*Proof.* The goal follows from $\mathrm{leadingdepthtwoblack}(Q, t)$. □

**Theorem 3.87.** *For every natural number $Q$ and natural number $b$ and natural number $r$ and natural number $g$, if $0 < r$, then if* $\mathrm{centerwindowagree}(b, b{+}Q, r + 1)$*, then if* $\mathrm{center}(b) = \mathrm{true}$*, then if* $\mathrm{whitecentergap}(b, r)$*, then if* $\mathrm{erasureevent}(Q, b + r, 0)$*, then if* $\mathrm{gatecode}(Q, b + r, 0) = g$*, then there exists a natural number $k$ such that $k \leq r + 1$ and* $\mathrm{leadingadvance}(Q, b, k) \wedge \mathrm{firstpatherasure}(Q, b, k, k - 2) \wedge \mathrm{gatecode}(Q, b + (k - 1), 0) = g$.

*Proof.* The goal follows from blackgaperasureorigin$(Q, b, r, g)$. □

**Theorem 3.88.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$ and natural number $g$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $g = 2$ or $g = 5$, then not not there exists a natural number $b$ such that there exists a natural number $k$ such that $N \leq b$ and $k \leq P \wedge$* center$(b) =$ true $\wedge$ whitecentergap$(b, k-1) \wedge$ leadingadvance$(m \cdot P, b, k) \wedge$ firstpatherasure$(m \cdot P, b, k, k-2) \wedge$ gatecode$(m \cdot P, b + (k-1), 0) = g$.

*Proof.* The goal follows from periodicblackerasureorigin$(T, P, m, N, g)$. □

**Theorem 3.89.** *For every natural number $t$,* rightcell$(t+1, 1) =$ bxor(center$(t)$, bxor(rightcell$(t, 1)$, right

*Proof.* The goal follows from rightimportstep$(t)$. □

**Theorem 3.90.** *For every natural number $Q$ and natural number $t$, if* center$(t+Q) =$ center$(t)$*, then* rightdefect$(Q, t+1, 1) =$ bxor(rightdefect$(Q, t, 1)$, rightimportdifference$(Q, t)$).

*Proof.* The goal follows from rightdefectimportbalance$(Q, t)$. □

**Theorem 3.91.** *For every natural number $Q$ and natural number $t$, if* center$(t) =$ true*, then if* center$(t + Q) =$ true*, then if* rightdefect$(Q, t, 1) =$ true*, then if* rightdefect$(Q, t+1, 1) =$ true*, then* rightphase$(Q, t+1, 1) = -$ rightphase$(Q, t, 1)$.

*Proof.* The goal follows from blackresetsurvival$(Q, t)$. □

**Theorem 3.92.** *For every natural number $Q$ and natural number $t$, if* center$(t) =$ true*, then if* center$(t + Q) =$ true*, then if* rightdefect$(Q, t, 1) =$ false*, then if* rightdefect$(Q, t + 1, 1) =$ true*, then* rightcell$(t, 1) =$ false *and* rightcell$(t + Q, 1) =$ false $\wedge$ rightdefect$(Q, t, 2) =$ true $\wedge$ rightphase$(Q, t+1, 1) = -$ rightphase$(Q, t, 2)$.

*Proof.* The goal follows from blackresetimport$(Q, t)$. □

**Theorem 3.93.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$ and truth value $before$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then not not there exists a natural number $t$ such that $N \leq t$ and* rightimport$(t) = before \wedge$ rightimport$(t + m \cdot P) =$ not$(before) \wedge$ rightdefect$(m \cdot P, t + 1, 1) =$ not(rightdefect$(m \cdot P, t, 1)$).

*Proof.* The goal follows from periodicrightimportorientation$(T, P, m, N, before)$. □

**Theorem 3.94.** *For every natural number $Q$ and natural number $t$, if* $\mathrm{center}(t+Q) = \mathrm{center}(t)$*, then if* $\mathrm{rightdefect}(Q,t,1) = \mathrm{false}$*, then if* $\mathrm{rightdefect}(Q,t+1,1) = \mathrm{true}$*, then* $\mathrm{rightcell}(t,1) = \mathrm{false}$ *and* $\mathrm{rightcell}(t+Q,1) = \mathrm{false} \wedge \mathrm{rightdefect}(Q,t,2) = \mathrm{true} \wedge \mathrm{rightimportdifference}(Q,t) = \mathrm{true} \wedge \mathrm{rightphase}(Q,t+1,1) = (1-2\cdot \mathrm{bitvalue}(\mathrm{center}(t)))\cdot \mathrm{rightphase}(Q,t,2)$.

*Proof.* The goal follows from $\mathrm{rightheadbirth}(Q,t)$. □

**Theorem 3.95.** *For every natural number $Q$ and natural number $start$ and natural number $finish$, if* $\mathrm{headimportpath}(Q,start,finish)$*, then if* $\mathrm{centercleanthrough}(Q,start,finish-start)$*, then* $\mathrm{rightphase}(Q,finish,1) = \mathrm{centersign}(start,finish-start)\cdot \mathrm{rightphase}(Q,start,2)$.

*Proof.* The goal follows from $\mathrm{headimportpathphase}(Q,start,finish)$. □

**Theorem 3.96.** *For every natural number $Q$ and natural number $t$, if* $\mathrm{leftdefect}(Q,t,0) = \mathrm{false}$*, then if* $\mathrm{leftdefect}(Q,t+1,0) = \mathrm{false}$*, then if* $\mathrm{center}(t) = \mathrm{false}$*, then* $\mathrm{leftphase}(Q,t,1) = (1-2\cdot \mathrm{bitvalue}(\mathrm{center}(t+1)))\cdot \mathrm{rightphase}(Q,t,1)$.

*Proof.* The goal follows from $\mathrm{whitecenterphasematching}(Q,t)$. □

**Theorem 3.97.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$ and natural number $g$, if $0 < P$, then if $0 < m$, then if* $\mathrm{centerperiodic}(T,P)$*, then if $g = 2$ or $g = 5$, then not not there exists a natural number $start$ such that there exists a natural number $b$ such that there exists a natural number $k$ such that $N \leq start$ and $N \leq b \wedge k \leq P \wedge \mathrm{center}(b) = \mathrm{true} \wedge \mathrm{whitecentergap}(b,k-1) \wedge \mathrm{leadingadvance}(m\cdot P,b,k) \wedge \mathrm{firstpatherasure}(m\cdot P,b,k,k-2) \wedge \mathrm{gatecode}(m\cdot P,b+(k-1),0) = g \wedge \mathrm{headimportpath}(m\cdot P,start,b+(k-1)) \wedge \mathrm{leftphase}(m\cdot P,b,k) + \mathrm{pathflux}(m\cdot P,b,k,k-1) = (1-2\cdot \mathrm{bitvalue}(\mathrm{center}(b+k)))\cdot \mathrm{centersign}(start,b+(k-1)-start)\cdot \mathrm{rightphase}(m\cdot P,start,2)$.

*Proof.* The goal follows from $\mathrm{periodiccoupledcancellation}(T,P,m,N,g)$. □

**Theorem 3.98.** *For every natural number $Q$ and natural number $b$ and natural number $k$, if* $\mathrm{leadingadvance}(Q,b,k)$*, then if* $\mathrm{centercleanthrough}(Q,b,k)$*, then* $\mathrm{leftphase}(Q,b,k) + \mathrm{pathflux}(Q,b,k,k-1) = (1-2\cdot \mathrm{bitvalue}(\mathrm{center}(b+k)))\cdot \mathrm{leftgatephase}(Q,b+(k-1),0)$.

*Proof.* The goal follows from $\mathrm{leadingarrivalphaseeqgatephase}(Q,b,k)$. □

**Theorem 3.99.** *For every natural number $Q$ and natural number $start$ and natural number $finish$, if* $\mathrm{headimportpath}(Q,start,finish)$*, then if* $\mathrm{centercleanthrough}(Q,start,finish-start)$*, then if* $\mathrm{center}(finish) = \mathrm{false}$*, then* $\mathrm{leftgatephase}(Q,finish,0) = \mathrm{centersign}(start,finish-start)\cdot \mathrm{rightphase}(Q,start,2)$.

*Proof.* The goal follows from headimportpathgatephase$(Q, start, finish)$. □

**Theorem 3.100.** *For every natural number $Q$ and natural number start and natural number $finish$, if* headimportpath$(Q, start, finish)$*, then if* centercleanthrough$(Q, start, finish - start)$*, then if* center$(finish) =$ false*, then not* leftgatephase$(Q, finish, 0) = -($centersign$(start, finish - start) \cdot$ rightphase$(Q, start, 2))$.

*Proof.* The goal follows from headimportpathrejectsnegativegate$(Q, start, finish)$. □

**Theorem 3.101.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$ and natural number $g$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $g = 2$ or $g = 5$, then not not there exists a natural number start such that there exists a natural number $b$ such that there exists a natural number $k$ such that $N \leq start$ and $N \leq b \wedge k \leq P \wedge$* center$(b) =$ true $\wedge$ whitecentergap$(b, k-1) \wedge$ leadingadvance$(m \cdot P, b, k) \wedge$ firstpatherasure$(m \cdot P, b, k, k-2) \wedge$ gatecode$(m \cdot P, b + (k-1), 0) = g \wedge$ headimportpath$(m \cdot P, start, b + (k-1)) \wedge$ leftgatephase$(m \cdot P, b + (k-1), 0) =$ centersign$(start, b + (k-1) - start) \cdot$ rightphase$(m \cdot P, start, 2)$.

*Proof.* The goal follows from periodiccoupledgatephasesign$(T, P, m, N, g)$. □

**Theorem 3.102.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq N$, then* pathflux$(m \cdot P + m \cdot P, N, N + (m \cdot P + m \cdot P), N + (m \cdot P + m \cdot P)) = -1$ *and* pathflux$(m \cdot P, N, N + (m \cdot P + m \cdot P), N + (m \cdot P + m \cdot P)) = 0 \wedge$ pathflux$(m \cdot P, N + m \cdot P, N + (m \cdot P + m \cdot P), N + (m \cdot P + m \cdot P)) = -1$.

*Proof.* The goal follows from periodicdoubledseedfluxsplit$(T, P, m, N)$. □

**Theorem 3.103.** leadingadvance$(11, 4, 4)$ *and* centercleanthrough$(11, 4, 4) \wedge$ headimportpath$(11, 6, 7) \wedge$centercleanthrough$(11, 6, 1) \wedge$leftgatephase$(11, 7, 0) = 1 \wedge$ centersign$(6, 1) = 1 \wedge$ rightphase$(11, 6, 2) = 1$.

*Proof.* The goal follows from seedcoupledgatesign. □

**Theorem 3.104.** *For every natural number $Q$ and natural number $t$ and natural number $j$, if* rightdefect$(Q, t+1, j+1) =$ true*, then* rightdefect$(Q, t, j) =$ true *or* rightdefect$(Q, t, j+1) =$ true $\vee$ rightdefect$(Q, t, j+2) =$ true.

*Proof.* The goal follows from rightdefecthasancestor$(Q, t, j)$. □

**Theorem 3.105.** *For every natural number $Q$ and natural number $t$ and natural number $j$ and natural number $T$, if $1 \leq j$, then if $T \leq t$, then if* rightdefect$(Q, t, j) = \text{true}$*, then if* centercleanthrough$(Q, T, t - T)$*, then* rightlineage$(Q, t, j, T)$.

*Proof.* The goal follows from rightlineageofcleancenter$(Q, t, j, T)$. □

**Theorem 3.106.** *For every natural number $Q$ and natural number $t$ and natural number $k$ and natural number $T$, if $1 \leq k$, then if $T \leq t$, then if* leftdefect$(Q, t, k) = \text{true}$*, then if* centercleanthrough$(Q, T, t - T)$*, then* leftlineage$(Q, t, k, T)$.

*Proof.* The goal follows from leftlineageofcleancenter$(Q, t, k, T)$. □

**Theorem 3.107.** *For every natural number $Q$ and natural number $t$ and natural number $j$ and natural number $T$, if $T \leq t$, then if* rightlineage$(Q, t, j, T)$*, then there exists a natural number $i$ such that $1 \leq i$ and $i \leq j + (t - T) \wedge j \leq i + (t - T) \wedge$* rightdefect$(Q, T, i) = \text{true}$.

*Proof.* The goal follows from rightlineageatstart$(Q, t, j, T)$. □

**Theorem 3.108.** *For every natural number $Q$ and natural number $t$ and natural number $k$ and natural number $T$, if $T \leq t$, then if* leftlineage$(Q, t, k, T)$*, then there exists a natural number $i$ such that $1 \leq i$ and $i \leq k + (t - T) \wedge k \leq i + (t - T) \wedge$* leftdefect$(Q, T, i) = \text{true}$.

*Proof.* The goal follows from leftlineageatstart$(Q, t, k, T)$. □

**Theorem 3.109.** *For every natural number $Q$ and natural number $t$ and natural number $j$, if* rightlineage$(Q, t, j, 0)$*, then there exists a natural number $i$ such that $1 \leq i$ and $i \leq Q \wedge i \leq j + t \wedge j \leq i + t \wedge$* rightdefect$(Q, 0, i) = \text{true} \wedge$ rightcell$(Q, i) = \text{true} \wedge$ rightphase$(Q, 0, i) = 1$.

*Proof.* The goal follows from rightlineageseed$(Q, t, j)$. □

**Theorem 3.110.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* leftlineage$(Q, t, k, 0)$*, then there exists a natural number $i$ such that $1 \leq i$ and $i \leq Q \wedge i \leq k + t \wedge k \leq i + t \wedge$* leftdefect$(Q, 0, i) = \text{true} \wedge$ leftcell$(Q, i) = \text{true} \wedge$ leftphase$(Q, 0, i) = 1$.

*Proof.* The goal follows from leftlineageseed$(Q, t, k)$. □

**Theorem 3.111.** *For every natural number $Q$ and natural number $t$, if $0 < Q$, then there exists a natural number $y$ such that $y \leq t + Q$ and* $\mathrm{rightfront}(Q, t, y)$.

*Proof.* The goal follows from $\mathrm{existsrightfront}(Q, t)$. □

**Theorem 3.112.** *For every natural number $Q$ and natural number $t$ and natural number $y$ and natural number $z$, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$*, then if* $\mathrm{rightfront}(Q, t, y)$*, then if* $\mathrm{rightfront}(Q, t + 1, z)$*, then* $y \leq z + 1$.

*Proof.* The goal follows from $\mathrm{rightfrontlower}(Q, t, y, z)$. □

**Theorem 3.113.** *For every natural number $Q$ and natural number $t$ and natural number $y$, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$*, then if* $\mathrm{rightfront}(Q, t, y)$*, then if* $2 \leq y$*, then* $\mathrm{rightdefect}(Q, t + 1, y - 1) = \mathrm{not}(\mathrm{rightcell}(t, y - 1))$.

*Proof.* The goal follows from $\mathrm{rightfrontdescent}(Q, t, y)$. □

**Theorem 3.114.** *For every natural number $Q$ and natural number $t$ and natural number $y$ and natural number $z$, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$*, then if* $\mathrm{rightfront}(Q, t, y)$*, then if* $2 \leq y$*, then if* $\mathrm{rightfront}(Q, t + 1, z)$*, then if* $z + 1 \leq y$*, then* $z = y - 1$ *and* $\mathrm{rightcell}(t, y - 1) = \mathrm{false}$.

*Proof.* The goal follows from $\mathrm{rightfrontlockstep}(Q, t, y, z)$. □

**Theorem 3.115.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number $s$, if* $\mathrm{rightmostleft}(Q, b, k)$*, then if* $\mathrm{centercleanthrough}(Q, b, k)$*, then if* $s+1 < k$*, then if* $\mathrm{rightdefect}(Q, b+s, 1) = \mathrm{true}$*, then* $\mathrm{center}(b + s) = \mathrm{true}$.

*Proof.* The goal follows from $\mathrm{leadingheadblackbeforearrival}(Q, b, k, s)$. □

**Theorem 3.116.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number start, if* $3 \leq k$*, then if* $\mathrm{leadingadvance}(Q, b, k)$*, then if* $\mathrm{centercleanthrough}(Q, b, k)$*, then if* $\mathrm{whitecentergap}(b, k - 1)$*, then if* $\mathrm{headimportpath}(Q, start, b + (k - 1))$*, then* $start = b + (k - 2)$.

*Proof.* The goal follows from $\mathrm{coupledimportbornatarrival}(Q, b, k, start)$. □

**Theorem 3.117.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number $s$, if $0 < Q$, then if $3 \leq k$, then if* $\mathrm{leadingadvance}(Q, b, k)$*, then if* $\mathrm{centercleanthrough}(Q, b, k)$*, then if* $\mathrm{whitecentergap}(b, k-1)$*, then if* $1 \leq s$*, then if* $s+1 < k$*, then there exists a natural number $y$ such that* $\mathrm{rightdefect}(Q, b + s, 1) = \mathrm{false}$ *and* $\mathrm{rightfront}(Q, b + s, y) \wedge 2 \leq y \wedge y \leq k - s$.

*Proof.* The goal follows from coupledrightfrontmirrorat$(Q, b, k, s)$. □

**Theorem 3.118.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number $s$, if $0 < Q$, then if $3 \leq k$, then if* leadingadvance$(Q, b, k)$*, then if* centercleanthrough$(Q, b, k)$*, then if* whitecentergap$(b, k-1)$*, then if $1 \leq s$, then if $s + 2 < k$, then if* rightfront$(Q, b + s, k - s)$*, then* rightcell$(b + s, k - s - 1) =$ false *and* rightfront$(Q, b + (s + 1), k - (s + 1))$.

*Proof.* The goal follows from coupledwhitestaircaseat$(Q, b, k, s)$. □

**Theorem 3.119.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$ and natural number $g$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $g = 2$ or $g = 5$, then not not there exists a natural number start such that there exists a natural number $b$ such that there exists a natural number $k$ such that there exists a natural number $y$ such that $N \leq start$ and $N \leq b \wedge T \leq b \wedge T \leq start \wedge 2 \leq k \wedge k \leq P \wedge$* center$(b) =$ true $\wedge$ whitecentergap$(b, k-1) \wedge$ leadingadvance$(m \cdot P, b, k) \wedge$ gatecode$(m \cdot P, b+(k-1), 0) = g \wedge$ headimportpath$(m \cdot P, start, b+(k-1)) \wedge (3 \leq k \to start = b+(k-2)) \wedge (3 \leq k \to$ rightfront$(m \cdot P, b+1, y) \wedge 2 \leq y \wedge y \leq k-1) \wedge$ rightlineage$(m \cdot P, start, 2, T) \wedge$ leftlineage$(m \cdot P, b, k, T)$.

*Proof.* The goal follows from periodiccoupledimportancestry$(T, P, m, N, g)$. □

**Theorem 3.120.** leadingadvance$(14, 5, 3)$ *and* centercleanthrough$(14, 5, 3) \wedge$ whitecentergap$(5, 2) \wedge$center$(5) =$ true $\wedge$ rightfront$(14, 6, 2) \wedge$headimportpath$(14, 6, 7)$.

*Proof.* The goal follows from seedcoupledimportatarrival. □

**Theorem 3.121.** *For every natural number $t$ and natural number $k$ and natural number $n$,* bitvalue(raybit$(t, k, n)) =$ bitvalue(raybit$(t, k, 0))+$rayflux$(t, k, n)$.

*Proof.* The goal follows from raybitbalance$(t, k, n)$. □

**Theorem 3.122.** *For every natural number $Q$ and natural number $d$ and natural number $t$ and natural number $p$, if $d + p \leq t$, then* leftphase$(Q + Q, t + (t - d - p), p) =$ diagonalphase$(Q, d, t) +$ diagonalflux$(Q, d, t, p)$.

*Proof.* The goal follows from diagonalraybalance$(Q, d, t, p)$. □

**Theorem 3.123.** *For every natural number $Q$ and natural number $R$ and natural number $d$ and natural number $t$ and natural number $p$,* diagonalflux$(Q+R, d, t, p) =$ diagonalflux$(Q, d, t, p) +$ diagonalflux$(R, d, t + Q, p)$.

*Proof.* The goal follows from diagonalfluxlagadd$(Q, R, d, t, p)$. □

**Theorem 3.124.** *For every natural number $Q$ and natural number $z$ and natural number $t$, if* zerodiagonal$(z)$*, then if* doublesat$(Q, z)$*, then if* settledby$(z+1, t)$*, then* diagonalphase$(Q, z+1, t) = -1 \vee$ diagonalphase$(Q, z+1, t) = 1$ *and* diagonalphase$(Q, z+1, t+Q) = -$ diagonalphase$(Q, z+1, t) \wedge$ diagonalphase$(Q+Q, z+1, t) = 0$.

*Proof.* The goal follows from diagonalphasedoubles$(Q, z, t)$. □

**Theorem 3.125.** *For every natural number $Q$ and natural number $z$ and natural number $t$, if* zerodiagonal$(z)$*, then if* settledby$(z+2, t)$*, then* diagonalphase$(Q, z+3, t+1) = -$ diagonalphase$(Q, z+1, t)$.

*Proof.* The goal follows from diagonalresetphase$(Q, z, t)$. □

**Theorem 3.126.** *For every natural number $Q$ and natural number $z$ and natural number $t$, if* zerodiagonal$(z)$*, then if* settledby$(z+2, t)$*, then if* $z+3 \leq t$*, then* diagonalflux$(Q, z+3, t+1, 1) =$ diagonalflux$(Q, z+1, t, 1) + 2 \cdot$ diagonalphase$(Q, z+1, t)$.

*Proof.* The goal follows from diagonalresetfluxjump$(Q, z, t)$. □

**Theorem 3.127.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $d$ and natural number $t$, if* centerperiodic$(T, P)$*, then if* $d+1 \leq t$*, then if* $T \leq t+(t-d-1)$*, then if* center$(t+(t-d-1)) =$ false*, then* diagonalphase$(m \cdot P, d, t)$ + diagonalflux$(m \cdot P, d, t, 1) = (1 - 2 \cdot$ bitvalue(center$(t+(t-d-1)+1))) \cdot$ rightphase$(m \cdot P + m \cdot P, t+(t-d-1), 1)$.

*Proof.* The goal follows from periodicdiagonalwhiteheadbalance$(T, P, m, d, t)$. □

**Theorem 3.128.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $d$ and natural number $t$ and natural number $start$, if* centerperiodic$(T, P)$*, then if* $d+1 \leq t$*, then if* $T \leq start$*, then if* headimportpath$(m \cdot P + m \cdot P, start, t+(t-d-1))$*, then if* center$(t+(t-d-1)) =$ false*, then* diagonalphase$(m \cdot P, d, t)$ + diagonalflux$(m \cdot P, d, t, 1) = (1 - 2 \cdot$ bitvalue(center$(t + (t-d-1) + 1))) \cdot$ (centersign$(start, t + (t-d-1) - start) \cdot$ rightphase$(m \cdot P + m \cdot P, start, 2))$.

*Proof.* The goal follows from periodicdiagonalimportbalance$(T, P, m, d, t, start)$. □

**Theorem 3.129.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number start and natural number $q$, if $0 < Q$, then if $3 \leq k$, then if* leadingadvance$(Q, b, k)$*, then if* centercleanthrough$(Q, b, k)$*, then if* whitecentergap$(b, k-1)$*, then if* headimportpath$(Q, start, b+(k-1))$*, then if $1 \leq q$, then if $q + 1 < k$, then* rightfront$(Q, b + 1, k - 1)$ *and* rightcell$(b + 1, q) =$ false $\wedge$ rightcell$(b + 1 + Q, q) =$ false.

*Proof.* The goal follows from coupledrightfrontexact$(Q, b, k, start, q)$. □

**Theorem 3.130.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number start and natural number $i$, if $0 < Q$, then if $3 \leq k$, then if* leadingadvance$(Q, b, k)$*, then if* centercleanthrough$(Q, b, k)$*, then if* whitecentergap$(b, k - 1)$*, then if* headimportpath$(Q, start, b + (k - 1))$*, then if $1 \leq i$, then if $i + 1 < k$, then* rightfront$(Q, b + i, k - i)$ *and* rightcell$(b + i, k - i - 1) =$ false $\wedge$ rightcell$(b + i + Q, k - i - 1) =$ false.

*Proof.* The goal follows from coupledforcedwhitestaircase$(Q, b, k, start, i)$. □

**Theorem 3.131.** *For every natural number $Q$ and natural number $b$ and natural number start, if* leadingadvance$(Q, b, 2)$*, then if* centercleanthrough$(Q, b, 2)$*, then if* headimportpath$(Q, start, b+1)$*, then* center$(b) =$ true *and* center$(b+1) =$ false $\wedge$($start < b \wedge$ rightdefect$(Q, b, 1) =$ true $\wedge$ rightimportdifference$(Q, b) =$ false $\wedge$ leftphase$(Q, b, 2) = (1-2 \cdot$ bitvalue(center$(b+2))) \cdot$ rightphase$(Q, b, 1) \vee$ $start = b \wedge$ rightdefect$(Q, b, 1) =$ false $\wedge$ rightcell$(b, 1) =$ false $\wedge$ rightcell$(b + Q, 1) =$ false $\wedge$ rightdefect$(Q, b, 2) =$ true $\wedge$ rightimportdifference$(Q, b) =$ true $\wedge$ leftphase$(Q, b, 2) = (1-2 \cdot$ bitvalue(center$(b+2))) \cdot$ rightphase$(Q, b, 2))$.

*Proof.* The goal follows from depthtwocancellationcases$(Q, b, start)$. □

**Theorem 3.132.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number start, if* leadingadvance$(Q, b, k)$*, then if* centercleanthrough$(Q, b, k)$*, then if* whitecentergap$(b, k-1)$*, then if* headimportpath$(Q, start, b+(k-1))$*, then $start = b+(k-2)$ or $k = 2$ and $start < b \wedge$* rightdefect$(Q, b, 1) =$ true.

*Proof.* The goal follows from coupledimportbirthordepthtwosurvival$(Q, b, k, start)$. □

**Theorem 3.133.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number start and natural number $q$, if $0 < Q$, then if $3 \leq k$, then if* leadingadvance$(Q, b, k)$*, then if* centercleanthrough$(Q, b, k)$*, then if* center$(b) =$ true*, then if* whitecentergap$(b, k-1)$*, then if* headimportpath$(Q, start, b+$

$(k-1))$, *then if* $q+2 < k$, *then* rightcell$(b, q) =$ true *and* rightcell$(b+Q, q) =$ true.

*Proof.* The goal follows from coupledblackrootprefix$(Q, b, k, start, q)$. □

**Theorem 3.134.** leadingadvance$(4, 5, 2)$ *and* centercleanthrough$(4, 5, 2) \wedge$ headimportpath$(4, 2, 6) \wedge$rightdefect$(4, 5, 1) =$ true $\wedge$ rightimportdifference$(4, 5) =$ false.

*Proof.* The goal follows from seeddepthtwosurvival. □

**Theorem 3.135.** *For every natural number* $Q$ *and natural number* $t$, alignedbackground$(Q, t+1) =$ wittphi(alignedbackground$(Q, t)$).

*Proof.* The goal follows from alignedbackgroundstep$(Q, t)$. □

**Theorem 3.136.** *For every natural number* $Q$ *and natural number* $t$, aligneddefect$(Q, t+1) =$ defectstep(alignedbackground$(Q, t)$, aligneddefect$(Q, t)$).

*Proof.* The goal follows from aligneddefectstep$(Q, t)$. □

**Theorem 3.137.** *For every natural number* $Q$ *and natural number* $t$, rowbit(alignedbackground$(Q, t), 2 \cdot t + Q) =$ true.

*Proof.* The goal follows from backgroundtopbit$(Q, t)$. □

**Theorem 3.138.** *For every natural number* $Q$ *and natural number* $t$, *if* $0 < Q$, *then* rowbit(aligneddefect$(Q, t), 2 \cdot (t + Q)) =$ true.

*Proof.* The goal follows from aligneddefecttopbit$(Q, t)$. □

**Theorem 3.139.** *For every natural number* $Q$ *and natural number* $t$, *if* $0 < Q$, *then* twopow$(2 \cdot (t + Q)) \leq$ aligneddefect$(Q, t)$ *and* aligneddefect$(Q, t) <$ twopow$(2 \cdot (t + Q) + 1)$.

*Proof.* The goal follows from aligneddefectmagnitude$(Q, t)$. □

**Theorem 3.140.** *For every natural number* $Q$ *and natural number* $t$ *and natural number* $j$, *if* $2 \cdot (t+Q) < j$, *then* rowbit(aligneddefect$(Q, t), j) =$ false.

*Proof.* The goal follows from aligneddefectabovetop$(Q, t, j)$. □

**Theorem 3.141.** *For every natural number* $Q$ *and natural number* $t$ *and natural number* $j$, *if* $j < Q$, *then* rowbit(aligneddefect$(Q, t), j) =$ rowbit(wittrow$(t+Q), j)$.

*Proof.* The goal follows from aligneddefectlowprefix$(Q, t, j)$. □

**Theorem 3.142.** *For every natural number $b$ and natural number $d$,* defectstep$(b, d) =$ wittxor(wittlinear$(d)$, wittshl(wittxor(wittcarry$(d)$, crosscarry$(b, d)), 1)$).

*Proof.* The goal follows from defectsteplinear$(b, d)$. □

**Theorem 3.143.** *For every natural number $b$ and natural number $d$,* rowbit(defectstep$(b, d), 0) =$ rowbit$(d, 0)$.

*Proof.* The goal follows from defectsteplowbit$(b, d)$. □

**Theorem 3.144.** *For every natural number $b$ and natural number $d$,* vala(defectstep$(b, d)) =$ valb(wittxor(wittcarry$(d)$, crosscarry$(b, d))$) *and* valb(defectstep$(b, d)) =$ bxor(vala(wittxor(wittcarry($d$

*Proof.* The goal follows from defectstepvalcoordinates$(b, d)$. □

**Theorem 3.145.** *For every natural number $j$,* vala(wittshl$(1, j + 3)) =$ vala(wittshl$(1, j)$) *and* valb(wittshl$(1, j+3)) =$ valb(wittshl$(1, j)) \wedge ($vala(wittshl$(1, j)) =$ true $\vee$ valb(wittshl$(1, j)) =$ true).

*Proof.* The goal follows from edgevalcoordinates$(j)$. □

**Theorem 3.146.** *For every natural number $t$ and natural number $k$,* aligneddefect(twopow$(k), t) =$ wittxor(wittwin$(t,$ twopow$(k))$, wittxor(wittrow$(t)$, wittshl(wittrow$(t)$, twopow$(k)+$ twopow$(k))))$.

*Proof.* The goal follows from aligneddefectwindow$(t, k)$. □

**Theorem 3.147.** *For every natural number $t$ and natural number $k$, if $t <$* twopow$(k)$, *then* rowbit(aligneddefect(twopow$(k), t), 0) =$ true *and* rowbit(wittwin$(t,$ twopow$(k)), 0) =$ false $\wedge$ rowbit(aligneddefect(twopow$(k), t), 2\cdot(t+$twopow$(k))) =$ true $\wedge$ rowbit(wittwin$(t,$ twopow$(k)), 2\cdot$ $(t +$ twopow$(k))) =$ false $\wedge$ vala(wittwin$(t,$ twopow$(k))) =$ vala(wittrow$(t +$ twopow$(k))) \wedge$ valb(wittwin$(t,$ twopow$(k))) =$ valb(wittrow$(t+$twopow$(k))) \wedge$ rowbit(aligneddefect(twopow$(k), t), t+$twopow$(k)) =$ rowbit(wittwin$(t,$ twopow$(k)), t+$ twopow$(k))$.

*Proof.* The goal follows from dyadicwindowreadings$(t, k)$. □

**Theorem 3.148.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* centerperiodic$(T, P)$, *then if $T \leq t$, then* rowbit(wittphi(aligneddefect$(m\cdot P, t)), t+m\cdot P+1) =$ rowbit(crosscarry(alignedbackground$(m\cdot P, t)$, aligneddefect$(m \cdot P, t)), t + m \cdot P)$.

*Proof.* The goal follows from periodiccentercrosscarry$(T, P, m, t)$. □

**Theorem 3.149.** aligneddefect$(1,0) = 5$ *and* vala(aligneddefect$(1,0)) =$ false $\wedge$ valb(aligneddefect$(1,0)) =$ true $\wedge$ aligneddefect$(1,2) = 93 \wedge$vala(aligneddefect$(1,2)) =$ false $\wedge$ valb(aligneddefect$(1,2)) =$ false $\wedge$ rowbit(aligneddefect$(1,2),3) =$ true $\wedge$ aligneddefect$(4,0) =$ $385 \wedge$vala(aligneddefect$(4,0)) =$ false $\wedge$ valb(aligneddefect$(4,0)) =$ false $\wedge$ rowbit(aligneddefect$(4,0),4) =$ false.

*Proof.* The goal follows from seedvalcancellation. □

**Theorem 3.150.** *For every natural number $d$ and natural number $n$ and natural number $j$, if $j < n$ or $n{+}2 < j$, then* rowbit(wittxor$(d,$ centerflipmask$(n)), n) =$ not(rowbit$(d,n)$) *and* rowbit(wittxor$(d,$ centerflipmask$(n)), j) =$ rowbit$(d,j) \wedge$ vala(wittxor$(d,$ centerflipmask$(n))) =$ vala$(d) \wedge$valb(wittxor$(d,$ centerflipmask$(n))) =$ valb$(d)$.

*Proof.* The goal follows from centerflipmaskinvisible$(d,n,j)$. □

**Theorem 3.151.** *For every natural number $t$ and natural number $j$,* rowbit(observercode$(t), j) =$ leftcell$(t,j)$.

*Proof.* The goal follows from observercodereadout$(t,j)$. □

**Theorem 3.152.** *For every natural number $Q$ and natural number $t$,* observerdefect$(Q,t) =$ wittxor(observercode$(t+Q)$, observercode$(t)$).

*Proof.* The goal follows from observerdefectalignment$(Q,t)$. □

**Theorem 3.153.** *For every natural number $Q$ and natural number $t$ and natural number $j$,* rowbit(observerdefect$(Q,t), j) =$ rowbit(aligneddefect$(Q,t), t+$ $Q+j)$.

*Proof.* The goal follows from observerdefectreadout$(Q,t,j)$. □

**Theorem 3.154.** *For every natural number $t$,* vala(observercode$(t+1)) =$ bxor(center$(t)$, bxor(vala(wittcarry(observercode$(t))),$ observerinput$(t))$) *and* valb(observercode$(t{+}1)) =$ bxor(center$(t)$, valb(wittcarry(observercode$(t))))$.

*Proof.* The goal follows from observervalstep$(t)$. □

**Theorem 3.155.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* centerperiodic$(T,P)$*, then if $T \leq$ $t$, then* vala(observerdefect$(m{\cdot}P, t{+}1)) =$ bxor(vala(wittxor(wittcarry(observercode$(t+$ $m{\cdot}P))$, wittcarry(observercode$(t)))))$, leftdefect$(m{\cdot}P,t,1))$ *and* valb(observerdefect$(m{\cdot}$ $P, t{+}1)) =$ valb(wittxor(wittcarry(observercode$(t{+}m{\cdot}P))$, wittcarry(observercode$(t))))$.

*Proof.* The goal follows from periodicobservervalstep$(T,P,m,t)$. □

**Theorem 3.156.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq t$, then* rowbit(observerdefect$(m \cdot P, t), t + m \cdot P) =$ true *and* rowbit(observerdefect$(m \cdot P, t), 0) =$ false.

*Proof.* The goal follows from periodicobserveredges$(T, P, m, t)$. □

**Theorem 3.157.** observerdefect$(2, 0) = 7$ *and* vala(observerdefect$(2, 0)) =$ false $\wedge$ valb(observerdefect$(2, 0)) =$ false $\wedge$ rowbit(observerdefect$(2, 0), 0) =$ true $\wedge$ observerdefect$(4, 0) = 24 \wedge$ vala(observerdefect$(4, 0)) =$ true $\wedge$ valb(observerdefect$(4, 0)) =$ true $\wedge$ rowbit(observerdefect$(4, 0), 0) =$ false.

*Proof.* The goal follows from observerseedzeroval. □

# 4 Cyclotomic observer transport

**Theorem 4.1.** *For every natural number $r$ and natural number $a$,* qdivides$(r,$ wittlinpow$(r, a))$.

*Proof.* The goal follows from qdividesintro$(r, a)$. □

**Theorem 4.2.** *For every natural number $r$ and natural number $d$, if* qdivides$(r, d)$*, then there exists a natural number $a$ such that* wittlinpow$(r, a) = d$.

*Proof.* The goal follows from qdivideselim$(r, d)$. □

**Theorem 4.3.** *For every natural number $r$ and natural number $d$ and natural number $n$,* qdivides$(r,$ wittshl$(d, n))$ *if and only if* qdivides$(r, d)$.

*Proof.* The goal follows from qdividesshl$(r, d, n)$. □

**Theorem 4.4.** *For every natural number $r$ and natural number $d$, if $0 < r$, then if* qdivides$(r, d)$*, then* vala$(d) =$ false *and* valb$(d) =$ false.

*Proof.* The goal follows from qdividesval$(r, d)$. □

**Theorem 4.5.** *For every natural number $r$ and natural number $d$, if $d \neq 0$, then if* qdivides$(r, d)$*, then $2 \cdot r \leq$* binarydegree$(d)$.

*Proof.* The goal follows from qdividesdegreebound$(r, d)$. □

**Theorem 4.6.** *For every natural number $d$, if $d \neq 0$, then there exists a natural number $r$ such that* qdivides$(r, d)$ *and not* qdivides$(r + 1, d)$.

*Proof.* The goal follows from qorderexists$(d)$. □

**Theorem 4.7.** *For every natural number $r$ and natural number $d$, if $d \neq 0$, then* $\mathrm{qdivides}(r, d)$ *if and only if* $r \leq \mathrm{qorder}(d)$.

*Proof.* The goal follows from $\mathrm{qdividesifforder}(r, d)$. □

**Theorem 4.8.** *For every natural number $Q$ and natural number $t$,* $\mathrm{observersourceflux}(Q, t, 0) = 0$.

*Proof.* The goal follows from $\mathrm{observersourcefluxzero}(Q, t)$. □

**Theorem 4.9.** *For every natural number $Q$ and natural number $t$ and natural number $n$,* $\mathrm{observersourceflux}(Q, t, n+1) = \mathrm{wittxor}(\mathrm{wittlinear}(\mathrm{observersourceflux}(Q, t, n)), \mathrm{wittshl}(\mathrm{ob}$ $n), n + 1))$.

*Proof.* The goal follows from $\mathrm{observersourcefluxstep}(Q, t, n)$. □

**Theorem 4.10.** *For every natural number $Q$ and natural number $t$ and natural number $n$ and natural number $s$,* $\mathrm{observersourceflux}(Q, t, n + s) = \mathrm{wittxor}(\mathrm{wittlinpow}(s, \mathrm{observersourceflux}(Q, t, n)), \mathrm{wittshl}(\mathrm{observersourceflux}(Q, t+ n, s), n))$.

*Proof.* The goal follows from $\mathrm{observersourcefluxappend}(Q, t, n, s)$. □

**Theorem 4.11.** *For every natural number $Q$ and natural number $t$,* $\mathrm{observerqrank}(Q, t) = t + Q - 2 \cdot \mathrm{qorder}(\mathrm{observerdefect}(Q, t))$.

*Proof.* The goal follows from $\mathrm{observerqrankdefinition}(Q, t)$. □

**Theorem 4.12.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq t$, then* $\mathrm{wittshl}(\mathrm{observerdefect}(m \cdot P, t+1), 1) = \mathrm{wittxor}(\mathrm{wittlinear}(\mathrm{observerdefect}(m \cdot P, t)), \mathrm{wittshl}(\mathrm{observersource}(m \cdot P, t), 1))$.

*Proof.* The goal follows from $\mathrm{periodicobserverpolynomialstep}(T, P, m, t)$. □

**Theorem 4.13.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq t$, then* $\mathrm{observersource}(m \cdot P, t) = \mathrm{wittxor}(\mathrm{wittxor}(\mathrm{wittcarry}(\mathrm{observercode}(t+ m \cdot P)), \mathrm{wittcarry}(\mathrm{observercode}(t))), \mathrm{naturalbit}(\mathrm{leftdefect}(m \cdot P, t, 1)))$.

*Proof.* The goal follows from $\mathrm{periodicobserversource}(T, P, m, t)$. □

**Theorem 4.14.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$ and natural number $n$, if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq t$, then* $\mathrm{wittshl}(\mathrm{observerdefect}(m \cdot P, t+n), n) = \mathrm{wittxor}(\mathrm{wittlinpow}(n, \mathrm{observerdefect}(m \cdot P, t)), \mathrm{observersourceflux}(m \cdot P, t, n))$.

*Proof.* The goal follows from periodicobserverpolynomialwindow$(T, P, m, t, n)$. □

**Theorem 4.15.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$ and natural number $n$ and natural number $r$ and natural number $k$, if* centerperiodic$(T, P)$*, then if $T \leq t$, then if* qdivides$(r,$ observerdefect$(m \cdot P, t))$*, then if $k \leq n+r$, then* qdivides$(k,$ observerdefect$(m \cdot P, t+n))$ *if and only if* qdivides$(k,$ observersourceflux$(m \cdot P, t, n))$.

*Proof.* The goal follows from periodicobserverwindowdivisibility$(T, P, m, t, n, r, k)$. □

**Theorem 4.16.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$ and natural number $r$, if* centerperiodic$(T, P)$*, then if $T \leq t$, then if* qdivides$(r,$ observerdefect$(m \cdot P, t))$*, then* qdivides$(r+1,$ observerdefect$(m \cdot P, t+1))$ *if and only if* qdivides$(r+1,$ observersource$(m \cdot P, t))$.

*Proof.* The goal follows from periodicobserverqgain$(T, P, m, t, r)$. □

**Theorem 4.17.** *For every natural number $Q$ and natural number $t$, if $0 < Q$, then* $2 \cdot$ qorder(observerdefect$(Q, t)) \leq t + Q$.

*Proof.* The goal follows from observerqorderbound$(Q, t)$. □

**Theorem 4.18.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$ and natural number $n$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq t$, then if $0 < n$, then* observerqrank$(m \cdot P, t + n) <$ observerqrank$(m \cdot P, t)$ *if and only if* qdivides(qorder(observerdefect$(m \cdot P, t)) + n/2 + 1,$ observersourceflux$(m \cdot P, t, n))$.

*Proof.* The goal follows from periodicobserverqrankdrop$(T, P, m, t, n)$. □

**Theorem 4.19.** observerdefect$(5, 2) = 216$ *and* observerdefect$(5, 3) = 412 \wedge$ qdivides$(1,$ observerdefect$(5, 2)) \wedge \neg$ qdivides$(1,$ observerdefect$(5, 3)) \wedge$ center$(2) =$ false $\wedge$ center$(7) =$ false $\wedge$ center$(3) =$ true $\wedge$ center$(8) =$ true.

*Proof.* The goal follows from observerseedqloss. □

**Theorem 4.20.** *For every natural number $r$ and natural number $a$ and natural number $d$, if* wittlinpow$(r, a) = d$*, then $a \leq d$.*

*Proof.* The goal follows from qquotientbound$(r, a, d)$. □

**Theorem 4.21.** *For every natural number $d$ and natural number $r$, if* qdivides$(r+1, d)$*, then* qordersearch$(d, r+1) = r+1$.

*Proof.* The goal follows from qordersearchhit$(d, r)$. □

**Theorem 4.22.** *For every natural number $d$ and natural number $r$, if not* qdivides$(r+1, d)$*, then* qordersearch$(d, r+1) =$ qordersearch$(d, r)$.

*Proof.* The goal follows from qordersearchmiss$(d, r)$. □

**Theorem 4.23.** *For every natural number $d$,* qorder$(d) =$ qordersearch$(d,$ binarydegree$(d))$.

*Proof.* The goal follows from qorderdefinition$(d)$. □

**Theorem 4.24.** *For every natural number $r$ and natural number $d$, if* qdivides$(r, d)$*, then there exists a natural number $a$ such that $a \leq d$ and* wittlinpow$(r, a) = d$.

*Proof.* The goal follows from qdividesboundedwitness$(r, d)$. □

**Theorem 4.25.** *For every natural number $d$,* qordersearch$(d, 0) = 0$.

*Proof.* The goal follows from qordersearchzero$(d)$. □

**Theorem 4.26.** *For every natural number $a$ and natural number $y$,* wittlinpow(twopow$(a), y) =$ wittxor(wittxor$(y,$ wittshl$(y,$ twopow$(a))),$ wittshl$(y,$ twopow$(a)+$twopow$(a)))$.

*Proof.* The goal follows from qfrobeniustwopow$(a, y)$. □

**Theorem 4.27.** wittlinpow$(16, 1) =$ wittxor(wittxor$(1,$ wittshl$(1, 16)),$ wittshl$(1, 32))$ *and* wittlinpow$(16,$ wittxor$(1,$ wittshl$(1, 16))) =$ wittxor$(1,$ wittshl$(1, 48))$.

*Proof.* The goal follows from qsixteenpage. □

**Theorem 4.28.** *For every natural number $a$ and natural number $y$,* qdivides(twopow$(a),$ wittxor$(y,$ witts twopow$(a))))$.

*Proof.* The goal follows from qdividespageshifttwopow$(a, y)$. □

**Theorem 4.29.** *For every natural number $a$ and natural number $n$ and natural number $H$,* qdivides(twopow$(a),$ wittxor$(H,$ wittxor(pagefold$(3\cdot$twopow$(a), H, n),$ pagetail$(3\cdot$ twopow$(a), H, n))))$.

*Proof.* The goal follows from qdividespagefoldtwopow$(a, n, H)$. □

**Theorem 4.30.** *For every natural number $L$ and natural number $H$ and natural number $p$, if $0 < L$, then* page$(L,$ wittcarry$(H), p) =$ wittxor(wittcarry(page$(L, H, p)),$ naturalbit(pa

*Proof.* The goal follows from pagecarry$(L, H, p)$. □

**Theorem 4.31.** *For every natural number $L$ and natural number $H$ and natural number $n$, if* $0 < L$, *then* pagefold$(L,$ wittcarry$(H), n) =$ carryledger$(L, H, n)$.

*Proof.* The goal follows from carryfoldeqledger$(L, H, n)$. □

**Theorem 4.32.** *For every natural number $a$ and natural number $H$ and natural number $n$,* qdivides(twopow$(a)$, wittxor(wittcarry$(H)$, wittxor(carryledger(3$\cdot$ twopow$(a), H, n)$, pagetail$(3 \cdot$ twopow$(a),$ wittcarry$(H), n))))$.

*Proof.* The goal follows from carryledgercongruencetwopow$(a, H, n)$. □

**Theorem 4.33.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$ and natural number $a$, if* centerperiodic$(T, P)$, *then if* $T \leq t$, *then* qdivides(twopow$(a)$, wittxor(observersource$(m{\cdot}P, t)$, wittxor(wittxor(carryledger(3$\cdot$ twopow$(a)$, observercode$(t{+}m{\cdot}P), t{+}m{\cdot}P{+}1)$, carryledger(3$\cdot$twopow$(a)$, observercode$(t), t+$ $m \cdot P + 1)),$ naturalbit(leftdefect$(m \cdot P, t, 1)))))$.

*Proof.* The goal follows from periodicobserversourceledger$(T, P, m, t, a)$. □

**Theorem 4.34.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$ and natural number $r$ and natural number $a$, if* centerperiodic$(T, P)$, *then if* $T \leq t$, *then if* $r + 1 \leq$ twopow$(a)$, *then if* qdivides$(r,$ observerdefect$(m \cdot P, t))$, *then* qdivides$(r{+}1,$ observerdefect$(m \cdot$ $P, t{+}1))$ *if and only if* qdivides$(r{+}1,$ wittxor(wittxor(carryledger(3$\cdot$twopow$(a)$, observercode$(t+$ $m \cdot P), t + m \cdot P + 1),$ carryledger$(3 \cdot$ twopow$(a),$ observercode$(t), t + m \cdot P +$ $1)),$ naturalbit(leftdefect$(m \cdot P, t, 1))))$.

*Proof.* The goal follows from periodicobserverqgainledger$(T, P, m, t, r, a)$. □

**Theorem 4.35.** qdivides$(16,$ wittlinpow$(16, 1))$ *and* wittxor(wittphi(wittxor(2, wittlinpow$(16, 1))),$ witt wittxor(wittlinpow$(17, 1), 4) \wedge \neg$ qdivides(16, wittxor(wittphi(wittxor(2, wittlinpow$(16, 1))),$ wittphi$(2))$

*Proof.* The goal follows from pageresiduenotclosed. □

**Theorem 4.36.** *For every natural number $a$ and natural number $A0$ and natural number $A1$ and natural number $A2$,* qdivides(twopow$(a)$, wittxor(wittxor(wittxor($A0$, wittshl($A1$ twopow$(a))),$ wittxor(wittxor($A0, A2$), wittshl(wittxor($A1, A2$), twopow$(a)))))$.

*Proof.* The goal follows from qblockresidue$(a, A0, A1, A2)$. □

**Theorem 4.37.** *For every natural number $a$ and natural number $A0$ and natural number $A1$ and natural number $A2$, if $A0 <$ twopow(twopow($a$)), then if $A1 <$ twopow(twopow($a$)), then if $A2 <$ twopow(twopow($a$)), then* qdivides(twopow($a$), wittxor(wittxor($A0$, wittshl($A1$, twopow($a$))), wittshl($A2$, twopow($a$)+ twopow($a$)))) *if and only if $A0 = A2$ and $A1 = A2$.*

*Proof.* The goal follows from qblockchargezeroiff($a, A0, A1, A2$). □

**Theorem 4.38.** *For every natural number $a$ and natural number $A$, if $A <$* twopow(twopow($a$)), *then* wittcarry(wittxor(wittxor($A$, wittshl($A$, twopow($a$))), wittshl($A$, twopow($a$)+ twopow($a$)))) = wittxor(wittxor(wittxor(wittcarry($A$), wittshl(wittcarry($A$), twopow($a$))), wittshl(witt twopow($a$))), wittxor(wittshl(naturalbit(blockendpoint(twopow($a$), $A$)), twopow($a$)), wittshl(naturalbit twopow($a$)))).

*Proof.* The goal follows from repeatedblockcarry($a, A$). □

**Theorem 4.39.** *For every natural number $a$ and natural number $A$, if $A <$* twopow(twopow($a$)), *then* qdivides(twopow($a$), wittxor(wittcarry(wittxor(wittxor($A$, wittshl($A$, twopow twopow($a$)))), naturalbit(blockendpoint(twopow($a$), $A$))))).

*Proof.* The goal follows from repeatedblockcarrycongruence($a, A$). □

**Theorem 4.40.** *For every natural number $a$ and natural number $H$ and natural number $p$ and natural number $A$, if $A <$* twopow(twopow($a$)), *then if* page(3·twopow($a$), $H, p$) = wittxor(wittxor($A$, wittshl($A$, twopow($a$))), wittshl($A$, twopow($a$)+ twopow($a$))), *then* qdivides(twopow($a$), wittxor(page(3·twopow($a$), wittcarry($H$), $p$), wittxor(naturalbit twopow($a$), $H, p$))))).

*Proof.* The goal follows from pagecarryrepeated($a, H, p, A$). □

**Theorem 4.41.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* centerperiodic($T, P$), *then if $T \leq t$, then* observersource($m \cdot P, t$) = wittxor(wittxor(wittcarry(observerdefect($m \cdot P, t$)), crosscarry(observercode($t$), observerdefect($m \cdot P, t$))), naturalbit(leftdefect($m \cdot P, t, 1$))).

*Proof.* The goal follows from periodicobserversourcecrosscarry($T, P, m, t$). □

**Theorem 4.42.** *There exists a natural number $s$ such that there exists a natural number $s2$ such that* qdivides(16, wittxor($s, s2$)) *and not* qdivides(16, wittxor(wittphi($s$), wittphi($s2$)

*Proof.* The goal follows from residuenotcongruence. □

**Theorem 4.43.** *For every natural number $Q$ and natural number $t$ and natural number $j$,* $\mathrm{rowbit}(\mathrm{observerdefect}(Q, t), j) = \mathrm{leftdefect}(Q, t, j)$.

*Proof.* The goal follows from $\mathrm{observerdefectleftreadout}(Q, t, j)$. □

**Theorem 4.44.** *For every natural number $r$ and natural number $e$ and natural number $k$, if $e \neq 0$, then if* $\mathrm{qdivides}(r, \mathrm{wittshl}(e, k))$, *then* $k + 2 \cdot r \leq \mathrm{binarydegree}(\mathrm{wittshl}(e, k))$.

*Proof.* The goal follows from $\mathrm{qfrontdegreebudget}(r, e, k)$. □

**Theorem 4.45.** *For every natural number $n$ and natural number $e$ and natural number $o$, if $e \neq 0$, then if* $\mathrm{twopow}(n) \leq o$, *then if* $\mathrm{wittshl}(e, o) < \mathrm{twopow}(3 \cdot \mathrm{twopow}(n))$, *then not* $\mathrm{qdivides}(\mathrm{twopow}(n), \mathrm{wittshl}(e, o))$.

*Proof.* The goal follows from $\mathrm{qpageoffsetcharge}(n, e, o)$. □

**Theorem 4.46.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $0 < Q$, then if* $\mathrm{rightmostleft}(Q, t, k)$, *then* $k + 2 \cdot \mathrm{qorder}(\mathrm{observerdefect}(Q, t)) \leq t + Q$.

*Proof.* The goal follows from $\mathrm{observerfrontqbudget}(Q, t, k)$. □

**Theorem 4.47.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $0 < Q$, then if* $\mathrm{rightmostleft}(Q, t, k)$, *then* $k \leq \mathrm{observerqrank}(Q, t)$.

*Proof.* The goal follows from $\mathrm{observerfrontqrank}(Q, t, k)$. □

**Theorem 4.48.** *For every natural number $a$ and natural number $Q$ and natural number $t$ and natural number $p$ and natural number $o$, if* $\mathrm{twopow}(a) \leq o$, *then if* $o < 3 \cdot \mathrm{twopow}(a)$, *then if* $\mathrm{rightmostleft}(Q, t, 3 \cdot \mathrm{twopow}(a) \cdot p + o)$, *then not* $\mathrm{qdivides}(\mathrm{twopow}(a), \mathrm{page}(3 \cdot \mathrm{twopow}(a), \mathrm{observerdefect}(Q, t), p))$.

*Proof.* The goal follows from $\mathrm{observerfrontpagecharge}(a, Q, t, p, o)$. □

**Theorem 4.49.** *For every natural number $Q$ and natural number $t$, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$, *then* $\mathrm{leftdefect}(Q, t + 1, 0) = \mathrm{bxor}(\mathrm{leftdefect}(Q, t, 1), \mathrm{centergate}(Q, t))$.

*Proof.* The goal follows from $\mathrm{centerdefectcoupling}(Q, t)$. □

**Theorem 4.50.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* $\mathrm{centerperiodic}(T, P)$, *then if $T \leq t$, then* $\mathrm{leftdefect}(m \cdot P, t, 1) = \mathrm{centergate}(m \cdot P, t)$.

*Proof.* The goal follows from $\mathrm{periodiccentercoupling}(T, P, m, t)$. □

**Theorem 4.51.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* $\mathrm{centerperiodic}(T, P)$*, then if* $T \leq t$*, then if* $\mathrm{center}(t) = \mathrm{true}$*, then* $\mathrm{leftdefect}(m \cdot P, t, 1) = \mathrm{false}$.

*Proof.* The goal follows from $\mathrm{periodicblacknoarrival}(T, P, m, t)$. □

**Theorem 4.52.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* $\mathrm{centerperiodic}(T, P)$*, then if* $T \leq t$*, then if* $\mathrm{center}(t) = \mathrm{false}$*, then* $\mathrm{leftdefect}(m \cdot P, t, 1) = \mathrm{rightdefect}(m \cdot P, t, 1)$.

*Proof.* The goal follows from $\mathrm{periodicwhitemirror}(T, P, m, t)$. □

**Theorem 4.53.** *For every natural number $T$ and natural number $P$, if* $\mathrm{center}(T + P) = \mathrm{center}(T)$*, then if for every natural number $t$, if* $T \leq t$*, then* $\mathrm{leftdefect}(P, t, 1) = \mathrm{centergate}(P, t)$*, then* $\mathrm{centerperiodic}(T, P)$.

*Proof.* The goal follows from $\mathrm{centerperiodicofcoupling}(T, P)$. □

**Theorem 4.54.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* $\mathrm{centerperiodic}(T, P)$*, then if* $T \leq t$*, then* $\mathrm{observersource}(m \cdot P, t) = \mathrm{wittxor}(\mathrm{wittxor}(\mathrm{wittcarry}(\mathrm{observercode}(t + m \cdot P)), \mathrm{wittcarry}(\mathrm{observercode}(t))), \mathrm{naturalbit}(\mathrm{centergate}(m \cdot P, t)))$.

*Proof.* The goal follows from $\mathrm{periodicobserversourcecentergate}(T, P, m, t)$. □

**Theorem 4.55.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* $\mathrm{centerperiodic}(T, P)$*, then if* $T \leq t$*, then if* $\mathrm{center}(t) = \mathrm{true}$*, then* $\mathrm{observersource}(m \cdot P, t) = \mathrm{wittxor}(\mathrm{wittcarry}(\mathrm{observercode}(t + m \cdot P)), \mathrm{wittcarry}(\mathrm{observercode}(t)))$.

*Proof.* The goal follows from $\mathrm{periodicblackobserversource}(T, P, m, t)$. □

**Theorem 4.56.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* $\mathrm{centerperiodic}(T, P)$*, then if* $T \leq t$*, then if* $\mathrm{center}(t) = \mathrm{false}$*, then* $\mathrm{observersource}(m \cdot P, t) = \mathrm{wittxor}(\mathrm{wittxor}(\mathrm{wittcarry}(\mathrm{observercode}(t + m \cdot P)), \mathrm{wittcarry}(\mathrm{observercode}(t))), \mathrm{naturalbit}(\mathrm{rightdefect}(m \cdot P, t, 1)))$.

*Proof.* The goal follows from $\mathrm{periodicwhiteobserversource}(T, P, m, t)$. □

**Theorem 4.57.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $b$ and natural number $k$ and natural number $s$, if* $\mathrm{centerperiodic}(T, P)$*, then if* $T \leq b$*, then if* $\mathrm{rightmostleft}(m \cdot P, b, k)$*, then if* $s + 1 < k$*, then* $\mathrm{leftdefect}(m \cdot P, b + s, 1) = \mathrm{false}$ *and* $\mathrm{observersource}(m \cdot P, b + s) = \mathrm{wittxor}(\mathrm{wittcarry}(\mathrm{observercode}(b + s + m \cdot P)), \mathrm{wittcarry}(\mathrm{observercode}(b + s)))$.

*Proof.* The goal follows from periodicfrontgapobserversource$(T, P, m, b, k, s)$. □

**Theorem 4.58.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $b$ and natural number $k$, if* centerperiodic$(T, P)$*, then if $T \leq b$, then if* rightmostleft$(m \cdot P, b, k)$*, then* leftdefect$(m \cdot P, b + (k - 1), 1) =$ true *and* center$(b+(k-1)) =$ false $\wedge$ rightdefect$(m \cdot P, b+(k-1), 1) =$ true $\wedge$ observersource$(m \cdot P, b+(k-1)) =$ wittxor(wittxor(wittcarry(observercode$(b+(k-1)+m \cdot P)),$ wittcarry(observercode$(b + (k - 1)))), 1)$.

*Proof.* The goal follows from periodicfrontarrivalobserversource$(T, P, m, b, k)$. □

**Theorem 4.59.** *For every natural number $Q$ and natural number start, if* center$(start) =$ false*, then if* center$(start+Q) =$ false*, then if* center$(start+1) =$ false*, then if* center$(start + 1 + Q) =$ false*, then if* rightcell$(start, 1) =$ false*, then if* rightcell$(start + Q, 1) =$ false*, then* leftgatephase$(Q, start + 1, 0) =$ rightphase$(Q, start, 2)$ *and* leftgatephase$(Q, start+1, 0) =$ centersign$(start, 1) \cdot$ rightphase$(Q, start, 2)$.

*Proof.* The goal follows from coupledarrivalgatephase$(Q, start)$. □

**Theorem 4.60.** *For every natural number $Q$ and natural number $t$, if $0 < Q$, then* rowbit(aligneddefect$(Q, t), 0) =$ true.

*Proof.* The goal follows from aligneddefectbitzero$(Q, t)$. □

**Theorem 4.61.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $b$ and natural number $k$, if* centerperiodic$(T, P)$*, then if $T \leq b$, then if* rightmostleft$(m \cdot P, b, k)$*, then* leftdefect$(m \cdot P, b + (k - 1), 1) =$ true *and* centergate$(m \cdot P, b + (k - 1)) =$ true $\wedge$ leftdefect$(m \cdot P, b + (k - 1) + 1, 0) =$ false.

*Proof.* The goal follows from periodicarrivalnocrossing$(T, P, m, b, k)$. □

**Theorem 4.62.** *For every natural number $Q$ and natural number $t$ and natural number $r$, if* leftadjacentdefect$(Q, t) =$ true*, then if* centercol$(t+r) =$ true*, then if* centercol$(t+r+Q) =$ true*, then* nearcollision$(Q, t, r+1, 2 \cdot (r + 1))$.

*Proof.* The goal follows from unitdefectforcesnearcollisionkit$(Q, t, r)$. □

**Theorem 4.63.** *For every natural number $t$,* center$(t) =$ centercol$(t)$.

*Proof.* The goal follows from centereqkit($t$). □

**Theorem 4.64.** *For every natural number $k$,* $0 <$ twopow($k$).

*Proof.* The goal follows from twopowpos($k$). □

**Theorem 4.65.** *For every natural number $b$ and natural number $d$,* defectstep($b, d$) = wittxor(wittphi(wittxor($b, d$)), wittphi($b$)).

*Proof.* The goal follows from defectstepeq($b, d$). □

**Theorem 4.66.** *For every natural number $y$ and natural number $Q$,* wittphi(wittshl($y, Q$)) = wittshl(wittphi($y$), $Q$).

*Proof.* The goal follows from phishl($y, Q$). □

**Theorem 4.67.** *For every natural number $Q$ and natural number $t$,* wittxor(alignedbackground($Q, t$), ali wittrow($t + Q$).

*Proof.* The goal follows from reconstruct($Q, t$). □

**Theorem 4.68.** *For every natural number $T$ and natural number $P$, if* $T \leq 512$*, then if* $0 < P$*, then if* $P \leq 128$*, then there exists a natural number $t$ such that* $T \leq t$ *and* $t < T + 32 \wedge \neg$ center($t + P$) = center($t$).

*Proof.* The goal follows from horizonthirtytwo($T, P$). □

**Theorem 4.69.** *For every natural number $T$ and natural number $P$, if* $T \leq 512$*, then if* $0 < P$*, then if* $P \leq 128$*, then not* centerperiodic($T, P$).

*Proof.* The goal follows from nocenterperiod512128($T, P$). □

**Theorem 4.70.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$ and natural number $k$, if* centerperiodic($T, P$)*, then if* $T \leq t$*, then if* leftdefect($m \cdot P, t, k$) = true*, then there exists a natural number $s$ such that* $t \leq s$ *and* $s < t + k \wedge$ rightdefect($m \cdot P, s, 1$) = true.

*Proof.* The goal follows from leftdefectsourced($T, P, m, t, k$). □

**Theorem 4.71.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$ and natural number $k$, if* centerperiodic($T, P$)*, then if* $T \leq t$*, then if* rightagrees($m \cdot P, t, k$)*, then* leftdefect($m \cdot P, t, k$) = false.

*Proof.* The goal follows from leftconeofrightagrees($T, P, m, t, k$). □

**Theorem 4.72.** *For every natural number $Q$ and natural number $t$ and integer $p$, if* $\mathrm{defect}(Q, t+1, p) = \mathrm{false}$*, then if* $\mathrm{defect}(Q, t, p) = \mathrm{false}$*, then if* $\mathrm{defect}(Q, t, p+1) = \mathrm{false}$*, then* $\mathrm{defect}(Q, t, p-1) = \mathrm{false}$.

*Proof.* The goal follows from $\mathrm{defectleftofagree}(Q, t, p)$. □

**Theorem 4.73.** $\mathrm{shortperiodsfail}(512, 128, 32) = \mathrm{true}$.

*Proof.* The goal follows from certificate51212832. □

**Theorem 4.74.** *For every natural number $T0$ and natural number $P0$ and natural number $H$ and natural number $T$ and natural number $P$, if* $\mathrm{shortperiodsfail}(T0, P0, H) = \mathrm{true}$*, then if* $T \leq T0$*, then if* $0 < P$*, then if* $P \leq P0$*, then* $\mathrm{centerdisagreeswithin}(T, P, H) = \mathrm{true}$.

*Proof.* The goal follows from $\mathrm{checkerofcertificate}(T0, P0, H, T, P)$. □

**Theorem 4.75.** *For every natural number $T$ and natural number $P$ and natural number $H$, if* $\mathrm{centerdisagreeswithin}(T, P, H) = \mathrm{true}$*, then there exists a natural number $t$ such that $T \leq t$ and $t < T + H \wedge \neg\, \mathrm{center}(t + P) = \mathrm{center}(t)$*.

*Proof.* The goal follows from $\mathrm{disagreementofchecker}(T, P, H)$. □

**Theorem 4.76.** *For every natural number $T$ and natural number $P$ and natural number $H$, if* $\mathrm{centerdisagreeswithin}(T, P, H) = \mathrm{true}$*, then not* $\mathrm{centerperiodic}(T, P)$.

*Proof.* The goal follows from $\mathrm{notcenterperiodicofchecker}(T, P, H)$. □

**Theorem 4.77.** *For every natural number $Q$ and natural number $t$,* $\mathrm{rightdefect}(Q, t, 1) = \mathrm{defect}(Q, t, 1)$.

*Proof.* The goal follows from $\mathrm{rightdefecteqdefect}(Q, t)$. □

**Theorem 4.78.** *For every natural number $t$,* $\mathrm{center}(t) = \mathrm{not}(\mathrm{ledgerdiagonal}(t))$.

*Proof.* The goal follows from $\mathrm{centereqnotledgerdiagonal}(t)$. □

**Theorem 4.79.** *For every natural number $T$ and natural number $P$,* $\mathrm{centerperiodic}(T, P)$ *if and only if* $\mathrm{ledgerperiodic}(T, P)$.

*Proof.* The goal follows from $\mathrm{centerperiodiciffledgerperiodic}(T, P)$. □

**Theorem 4.80.** *For every natural number $t$,* $\mathrm{wittrow}(t) = \mathrm{wittxor}(\mathrm{wittlinpow}(t, 1), \mathrm{wittwin}(0, t))$.

*Proof.* The goal follows from $\mathrm{roweqlinearxorledger}(t)$. □

**Theorem 4.81.** *For every natural number $y$,* $\mathrm{wittphi}(y) = \mathrm{wittxor}(\mathrm{wittlinear}(y), \mathrm{wittshl}(\mathrm{wittcarry}(y), 1$

*Proof.* The goal follows from $\mathrm{phieqlinearxorcarry}(y)$. □

**Theorem 4.82.** *For every natural number $a$ and natural number $b$,* $\mathrm{wittlinear}(\mathrm{wittxor}(a, b)) = \mathrm{wittxor}(\mathrm{wittlinear}(a), \mathrm{wittlinear}(b))$.

*Proof.* The goal follows from $\mathrm{philinearxor}(a, b)$. □

**Theorem 4.83.** *For every natural number $m$,* $\mathrm{trailingparity}(2 \cdot m) = \mathrm{false}$.

*Proof.* The goal follows from $\mathrm{trailingparityeven}(m)$. □

**Theorem 4.84.** *For every natural number $m$,* $\mathrm{trailingparity}(2 \cdot m + 1) = \mathrm{not}(\mathrm{trailingparity}(m))$.

*Proof.* The goal follows from $\mathrm{trailingparityodd}(m)$. □

**Theorem 4.85.** *For every natural number $t$,* $\mathrm{rowbit}(\mathrm{philinpow}(1, t, 1), t) = \mathrm{true}$.

*Proof.* The goal follows from $\mathrm{linearcenter}(t)$. □

**Theorem 4.86.** *For every natural number $t$,* $\mathrm{rowbit}(\mathrm{philinpow}(1, t, 1), t + 1) = \mathrm{trailingparity}(t)$.

*Proof.* The goal follows from $\mathrm{linearabovecenter}(t)$. □

**Theorem 4.87.** *For every natural number $t$, if $1 \leq t$, then* $\mathrm{rowbit}(\mathrm{philinpow}(1, t, 1), t - 1) = \mathrm{trailingparity}(t)$.

*Proof.* The goal follows from $\mathrm{linearbelowcenter}(t)$. □

**Theorem 4.88.** *For every natural number $s$ and natural number $y$,* $\mathrm{philin}(s, \mathrm{philin}(s, y)) = \mathrm{philin}(2 \cdot s, y)$.

*Proof.* The goal follows from $\mathrm{philinphilin}(s, y)$. □

**Theorem 4.89.** *For every natural number $m$ and natural number $y$,* $\mathrm{philinpow}(1, 2 \cdot m, y) = \mathrm{philinpow}(2, m, y)$.

*Proof.* The goal follows from $\mathrm{philinpowtwomul}(m, y)$. □

**Theorem 4.90.** *For every natural number $m$ and natural number $y$,* $\mathrm{wittlinpow}(m, y) = \mathrm{philinpow}(1, m, y)$.

*Proof.* The goal follows from $\mathrm{philinearpoweqphilinpow}(m, y)$. □

**Theorem 4.91.** *For every natural number $t$, if $1 \leq t$, then* $\mathrm{center}(t+1) = \mathrm{rule}(\mathrm{bxor}(\mathrm{trailingparity}(t), \mathrm{ledgerbit}(t, t{+}1)), \mathrm{not}(\mathrm{ledgerbit}(t, t)), \mathrm{bxor}(\mathrm{trailingparity}(t), \mathrm{ledgerbit}(t, t{-}1)))$.

*Proof.* The goal follows from $\mathrm{centersuccledgerpaper}(t)$. □

**Theorem 4.92.** *For every natural number $j$ and natural number $t$,* $\mathrm{rdiag}(j, t) = \mathrm{bxor}(\mathrm{rowbit}(\mathrm{philinpow}(1, t, 1), j), \mathrm{ledgerbit}(t, j))$.

*Proof.* The goal follows from $\mathrm{rdiageqlinearbxorledger}(j, t)$. □

**Theorem 4.93.** *For every natural number $t$,* $\mathrm{cell}(t, -1) = \mathrm{bxor}(\mathrm{trailingparity}(t), \mathrm{ledgerbit}(t, t{+}1))$.

*Proof.* The goal follows from $\mathrm{celllefteqpaper}(t)$. □

**Theorem 4.94.** *For every natural number $t$, if $1 \leq t$, then* $\mathrm{cell}(t, 1) = \mathrm{bxor}(\mathrm{trailingparity}(t), \mathrm{ledgerbit}(t, t-1))$.

*Proof.* The goal follows from $\mathrm{cellrighteqpaper}(t)$. □

**Theorem 4.95.** *For every natural number $T$ and natural number $P$, if $0 < P$, then if* centerperiodic$(T, P)$*, then* centerkernelfinite.

*Proof.* The goal follows from $\mathrm{centerkernelfiniteofcenterperiodic}(T, P)$. □

**Theorem 4.96.** *For every natural number $T$ and natural number $P$, if* centerinfinitekernel*, then if $0 < P$, then not* centerperiodic$(T, P)$.

*Proof.* The goal follows from $\mathrm{notcenterperiodicofinfinitekernel}(T, P)$. □

**Theorem 4.97.** *If* centerinfinitekernel*, then* problemone.

*Proof.* Assume $h$. The goal follows from $\mathrm{problem1ofinfinitekernel}(h)$. □

**Theorem 4.98.** *For every natural number $k$ and natural number $r$ and natural number $t$,* $\mathrm{kernelseq}(k, r, t{+}1) = \mathrm{bxor}(\mathrm{bxor}(\mathrm{kernelseq}(k, r, t), \mathrm{cell}(\mathrm{twopow}(k) \cdot t + r, \mathrm{negpow}(k))), \mathrm{response}(\mathrm{twopow}(k) \cdot t + r, k))$.

*Proof.* The goal follows from $\mathrm{kernelseqstep}(k, r, t)$. □

**Theorem 4.99.** *For every natural number $t$ and natural number $k$, if $t <$* twopow$(k)$*, then* $\mathrm{response}(t, k) = \mathrm{bxor}(\mathrm{center}(t), \mathrm{center}(t + \mathrm{twopow}(k)))$.

*Proof.* The goal follows from $\mathrm{responsecone}(t, k)$. □

**Theorem 4.100.** *For every natural number $k$ and natural number $r$ and natural number $t$,* $\mathrm{kernelseq}(k+1, r, t) = \mathrm{kernelseq}(k, r, 2 \cdot t)$.

*Proof.* The goal follows from $\mathrm{kernelseqeven}(k, r, t)$. □

**Theorem 4.101.** *For every natural number $k$ and natural number $r$ and natural number $t$,* $\mathrm{kernelseq}(k+1, r+\mathrm{twopow}(k), t) = \mathrm{kernelseq}(k, r, 2 \cdot t+1)$.

*Proof.* The goal follows from $\mathrm{kernelseqodd}(k, r, t)$. □